MOTOR VALLEY
UNIVERSITY OF
EMILIA-ROMAGNA

ELECTRIC DRIVELINES

PORSCHE 930 TURBO HYBRIDISATION



Team Members

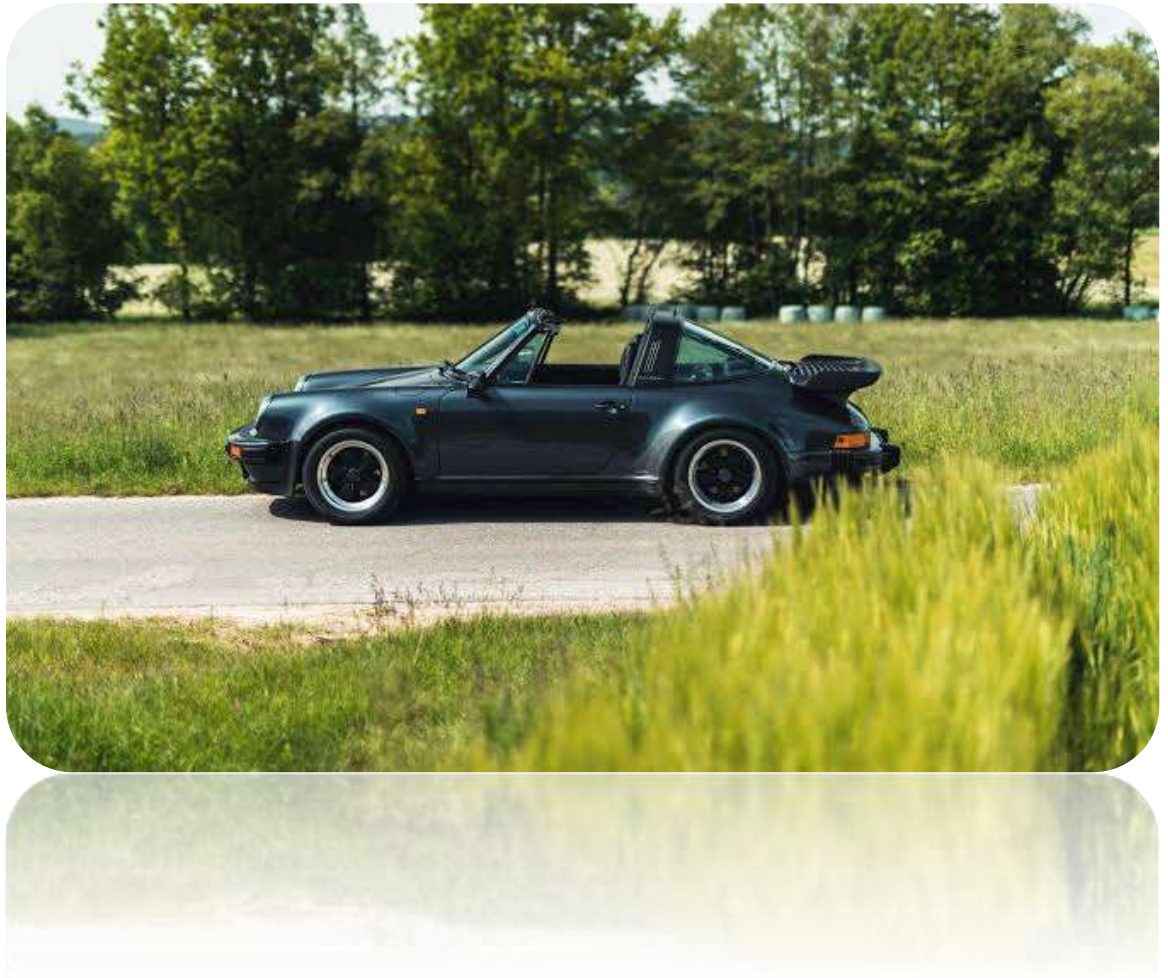
Advanced Automotive Engineering - Advanced Powertrain

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- Testi Riccardo

Electric Vehicle Engineering

- Jayakumar Satyasagar





The Goal

- Develop a street homologated version of the iconic *Porsche 930 Turbo* to **eliminate low-end turbo lag** and improve vehicle efficiency while preserving its classic character.

- **Baseline** — *1978 Porsche 930 Turbo 3.3L: Air-cooled Flat-6, 300 Hp, 412 Nm & 1300 kg*

- **Hybrid Powertrain Target:**
  367.5 Hp Peak, 500+ Nm of Torque,
  4-Mode Supervisory ECU & ~ 60 kg retrofit hardware.

"The Widowmaker"



The Model

Vehicle Information	Value	Units
Gross Vehicle Mass	1542.5	kg
Curb Weight	1300	kg
Front Mass Distribution	41.5	%
Wheelbase	2.272	m
Front distance to CoG	1.329	m
Rear distance to CoG	0.943	m
Height of CoG	0.48	m
Drag Coefficient	0.39	-
Frontal Area	1.77	m^2

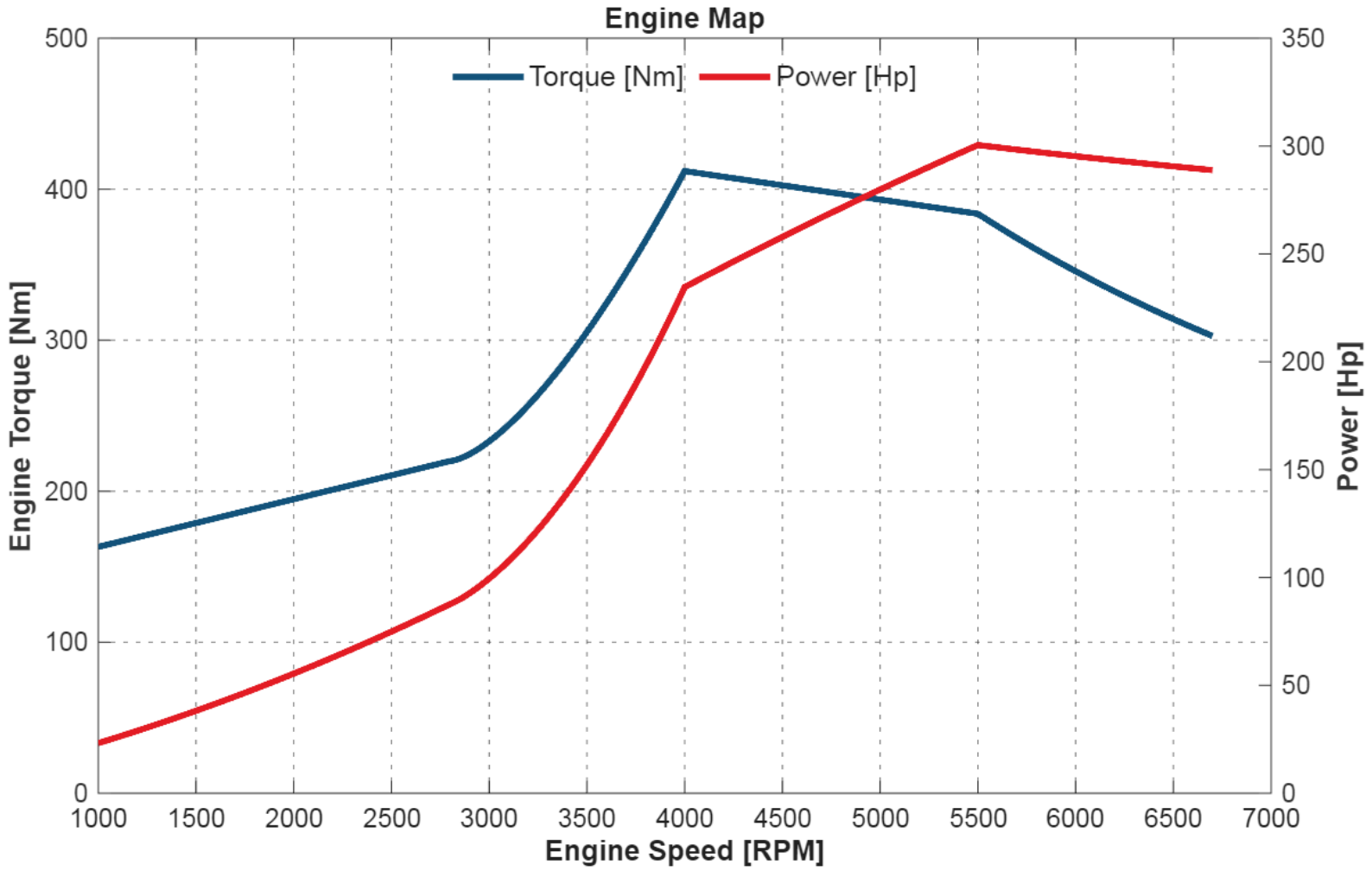


Chassis & Tires	Value	Units	Notes
Dynamic Tire Radius	0.314	m	Rear OEM 245/45 R16, dw = 16"
Rolling Resistance	0.015	-	Performance dry tire on asphalt
Road Friction Coeff (μ_{road})	0.90	-	Dry asphalt reference adhesion

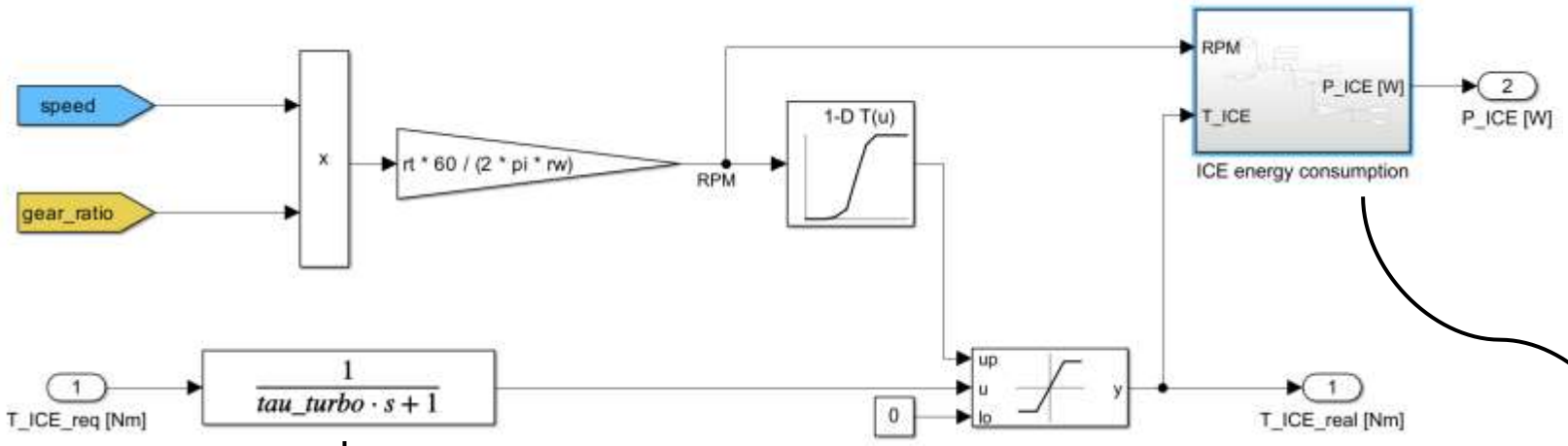
Powertrain Specifications

Stock Powertrain	Specification
ICE Architecture	3.3L Flat-6 Turbo
ICE Peak Power	221 kW @ 5500 rpm
ICE Peak Torque	412 Nm @ 4000 rpm
BSFC	395 g/kWh

Transmission	
1st Gear Ratio	2.25
2nd Gear Ratio	1.30
3rd Gear Ratio	0.89
4th Gear Ratio	0.625
Final Gear Reduction	4.22
Drive Type	RWD



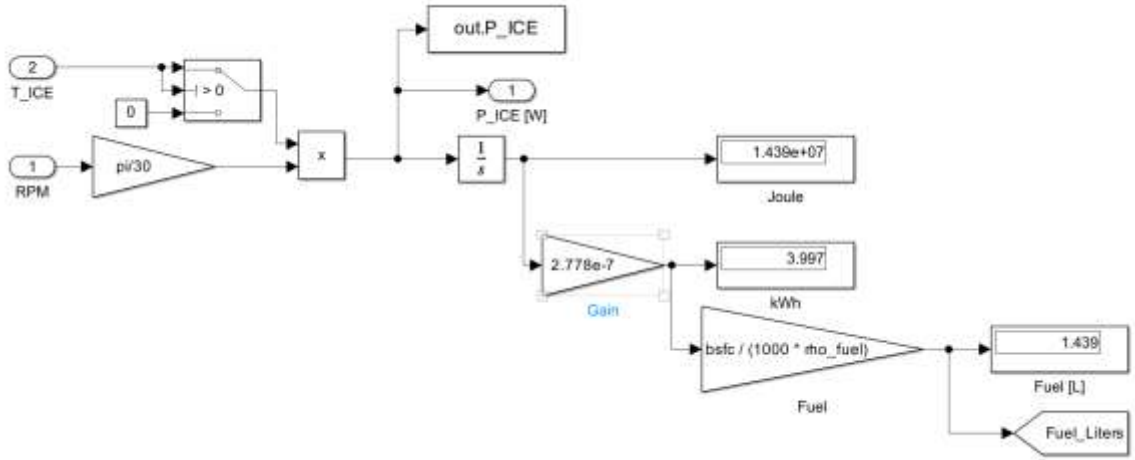
Stock ICE – Modelling





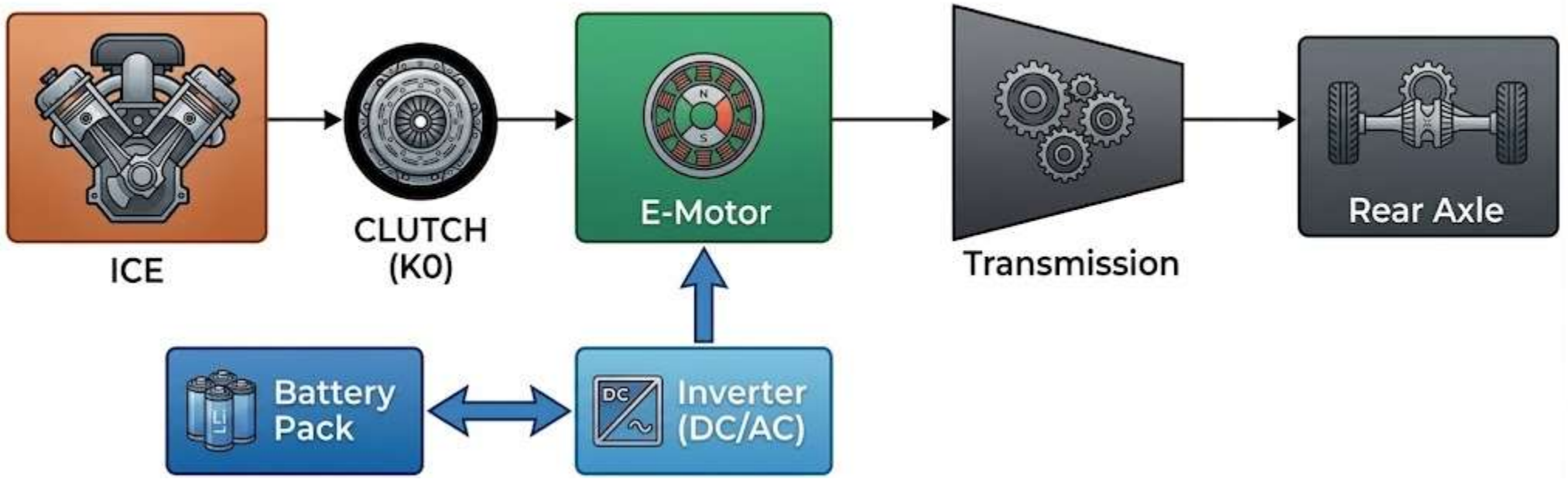


Porsche 930 Turbo K27 KKK (Kühnle, Kopp & Kausch)

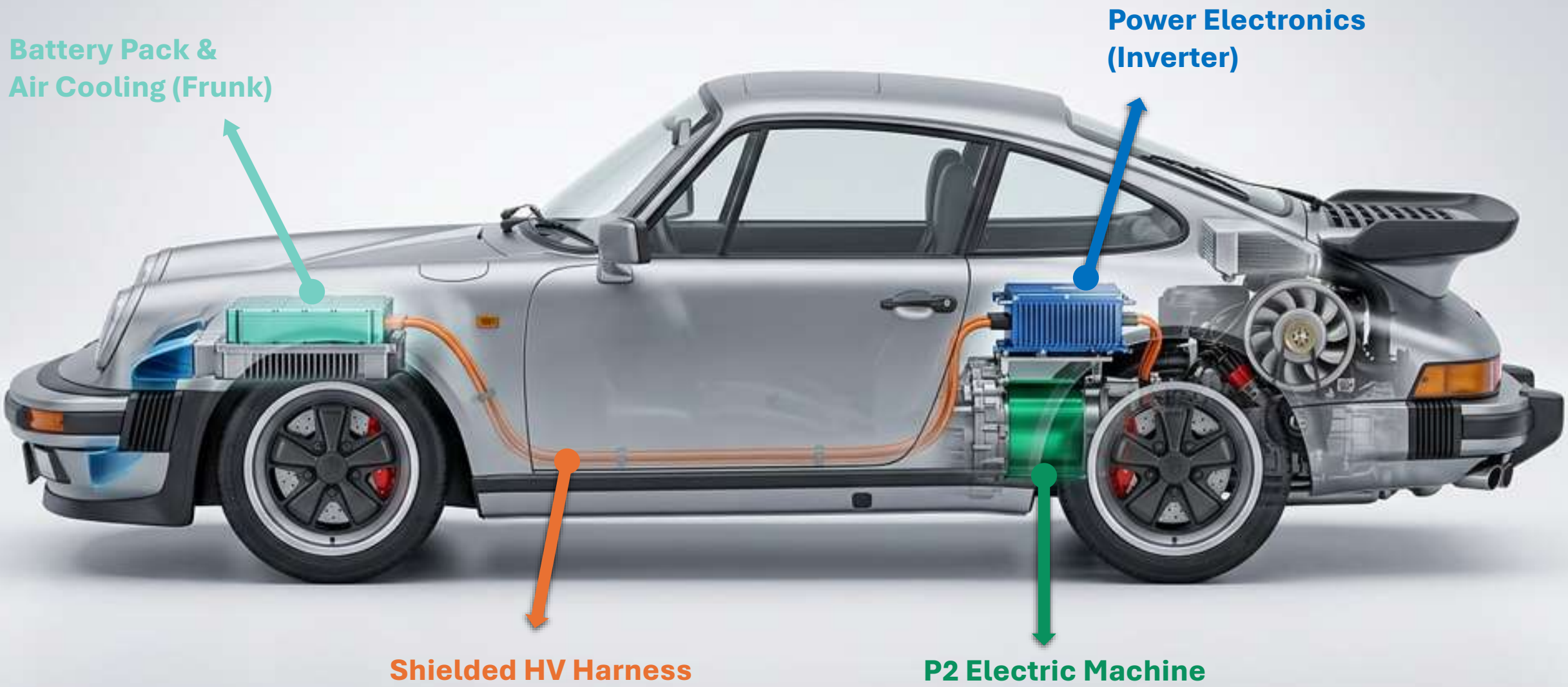


Hybrid System Architecture Definition

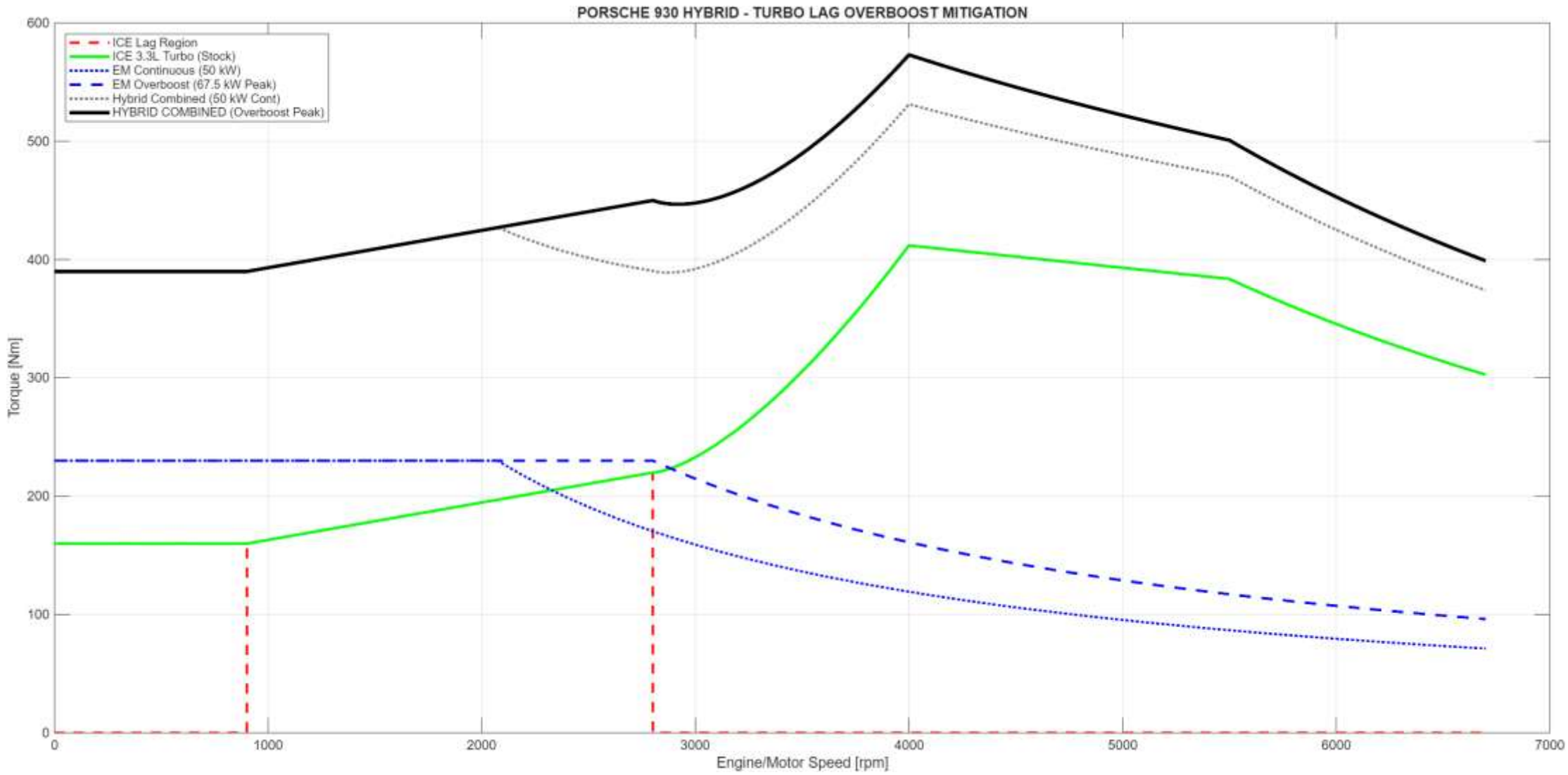
Parallel hybrid architecture with EM in **P2 configuration**



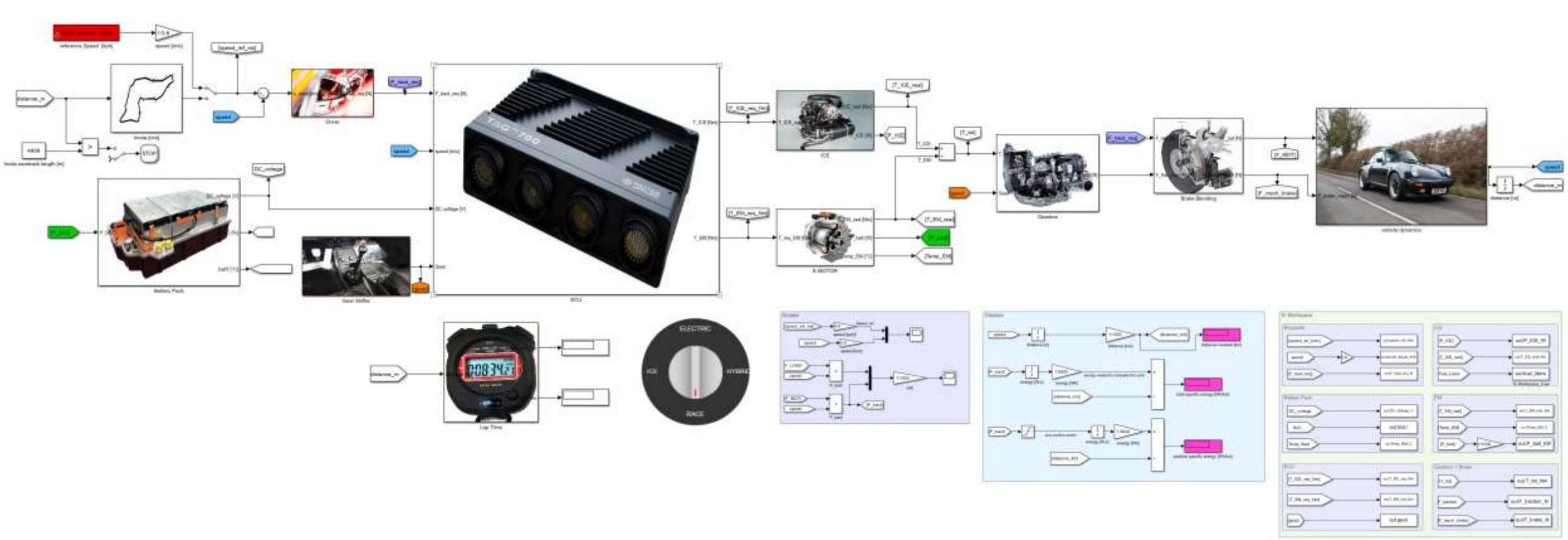
Overall drivetrain concept



Hybridization Potential



Simulink Scheme



Vehicle Dynamics

Aerodynamic drag:

$$F_{\text{aero}} = \frac{1}{2}\rho c_d a_f u^2$$

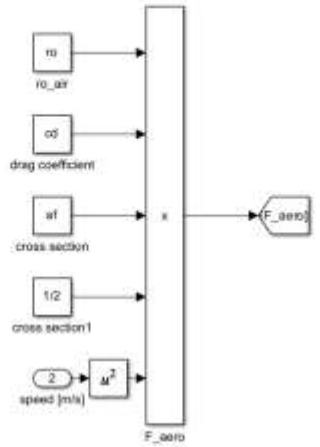
Rolling resistance:

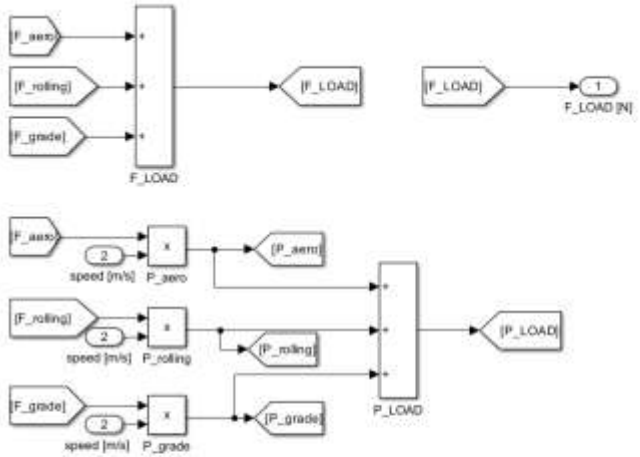
$$F_{rolling} = crr \cdot \cos(\alpha) \cdot 9.81 \cdot mg$$

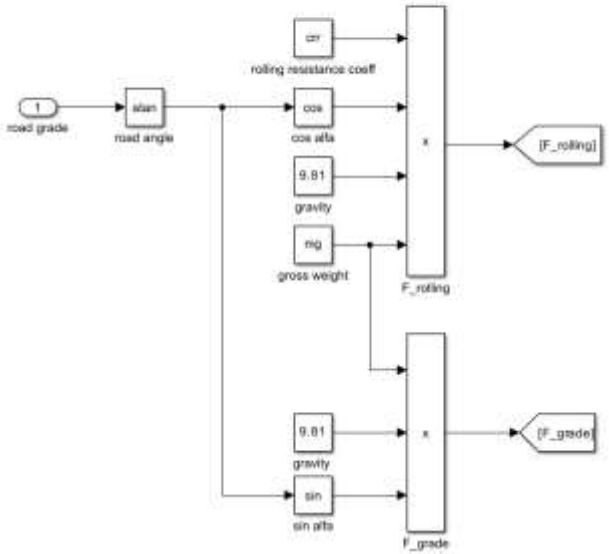
Gradient resistance:

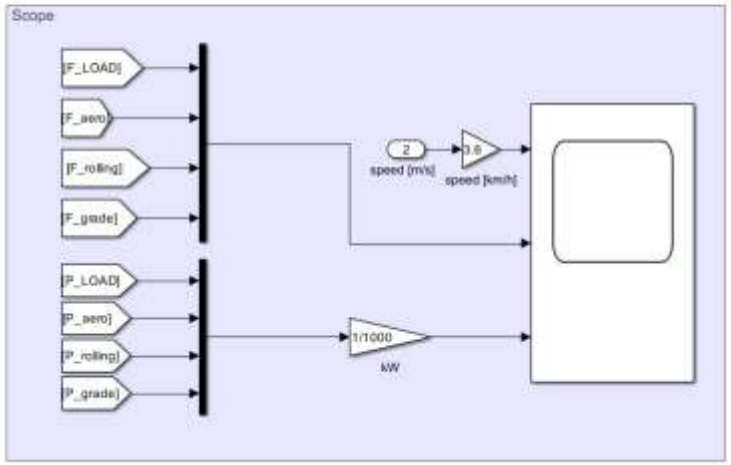
$$F_{grade} = mg \cdot 9.81 \cdot \sin(\alpha)$$

Aerodynamics	Value	Units
Drag Coefficient	0.39	-
Frontal Area	1.77	m²
Air Density (ρ @ 20°C)	1.204	kg/m³
Lift Coefficient	-0.05	-







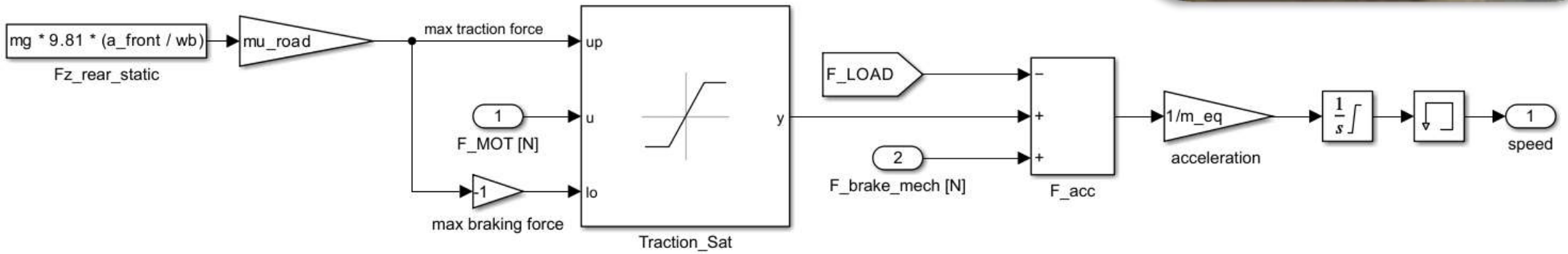


Vehicle Load Forces

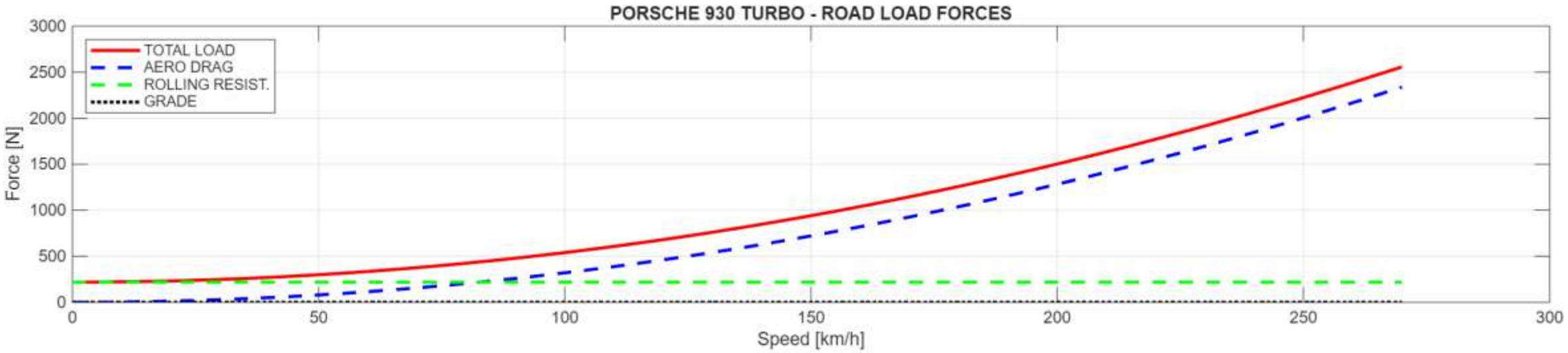
Maximum Traction Force:

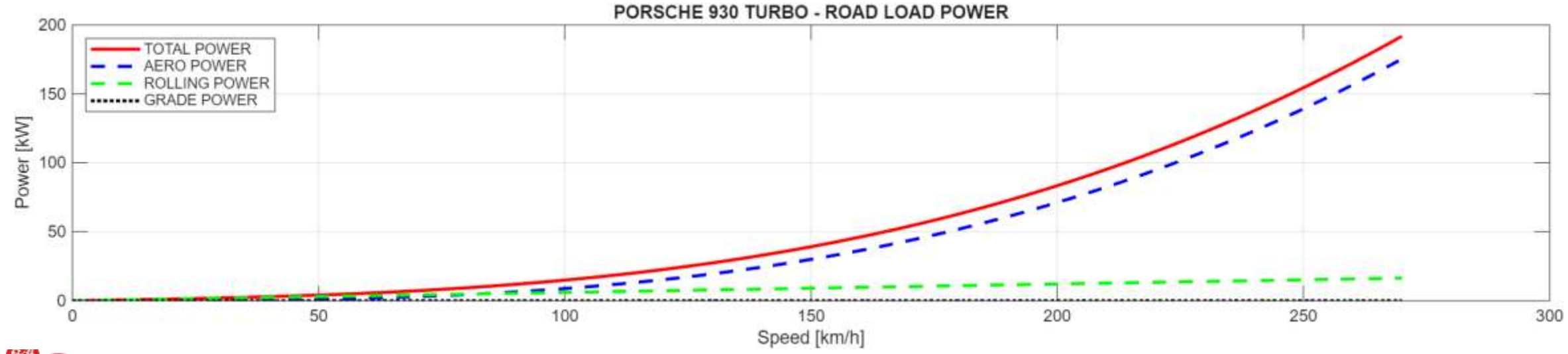
$$F_{z,\,rear,\,static} = mg \cdot 9.81 \cdot (1 - CoG) \cdot \mu_{road}$$



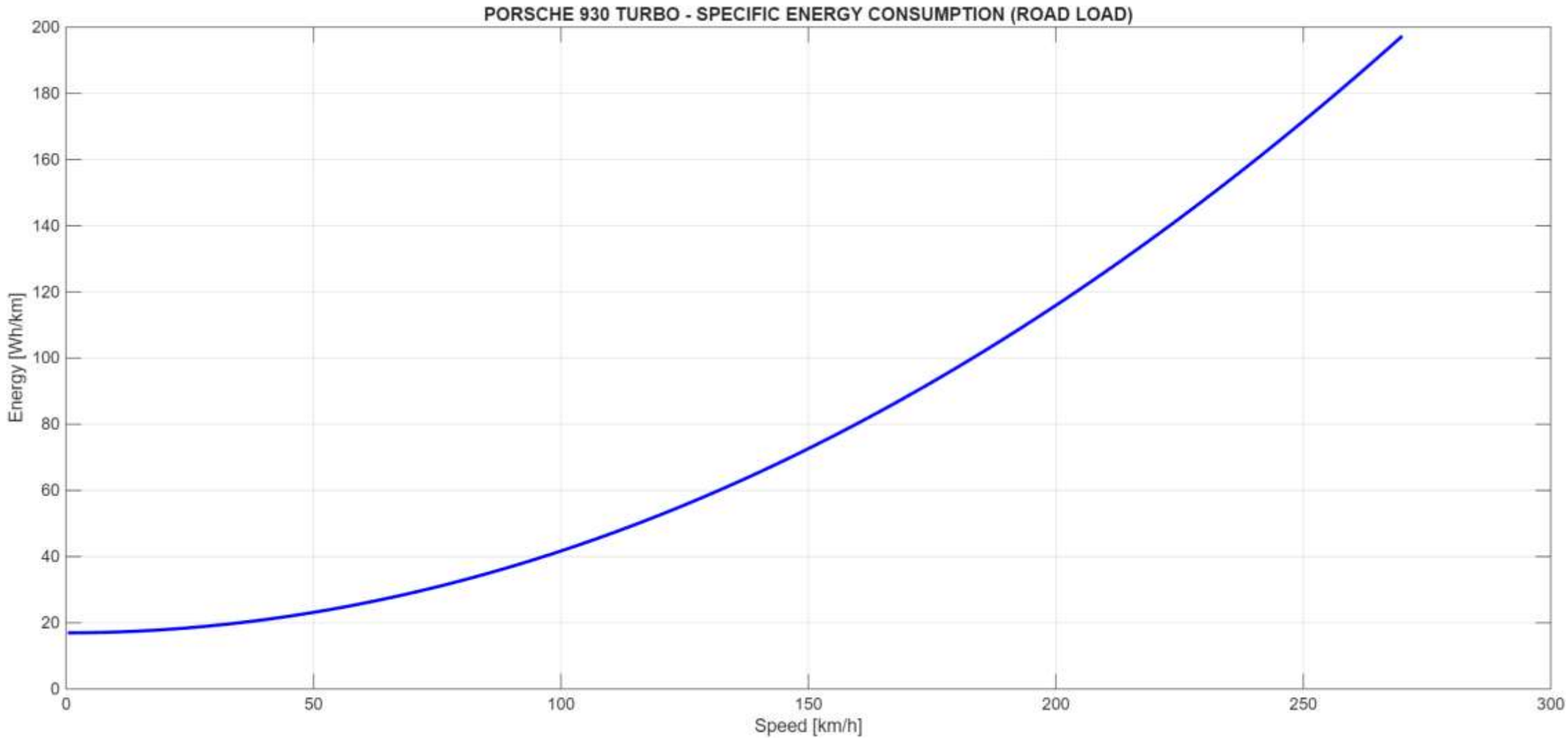


Road Loads


PORSCHE 930 TURBO - ROAD LOAD FORCES
TOTAL LOAD
AERO DRAG
ROLLING RESIST.
GRADE
Force [N]
Speed [km/h]
0
500
1000
1500
2000
2500
3000
0
50
100
150
200
250
300

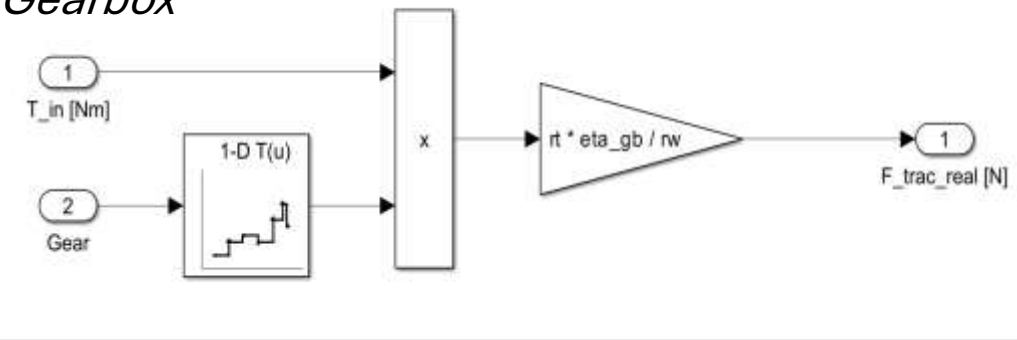

PORSCHE 930 TURBO - ROAD LOAD POWER
TOTAL POWER
AERO POWER
ROLLING POWER
GRADE POWER
Power [kW]
Speed [km/h]
0
50
100
150
200
0
50
100
150
200
250
300

Energy Consumption

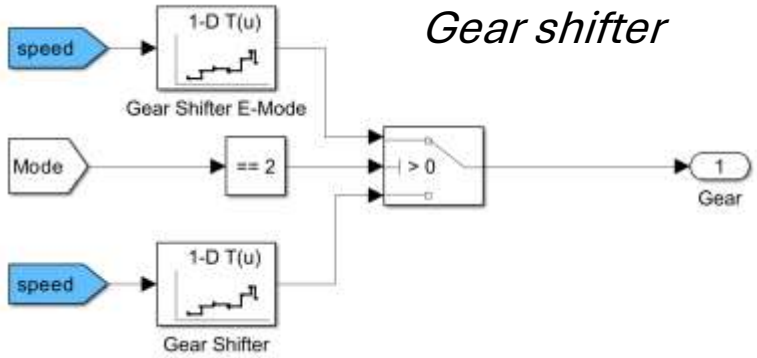


Gearbox & Gear Shifter

Gearbox

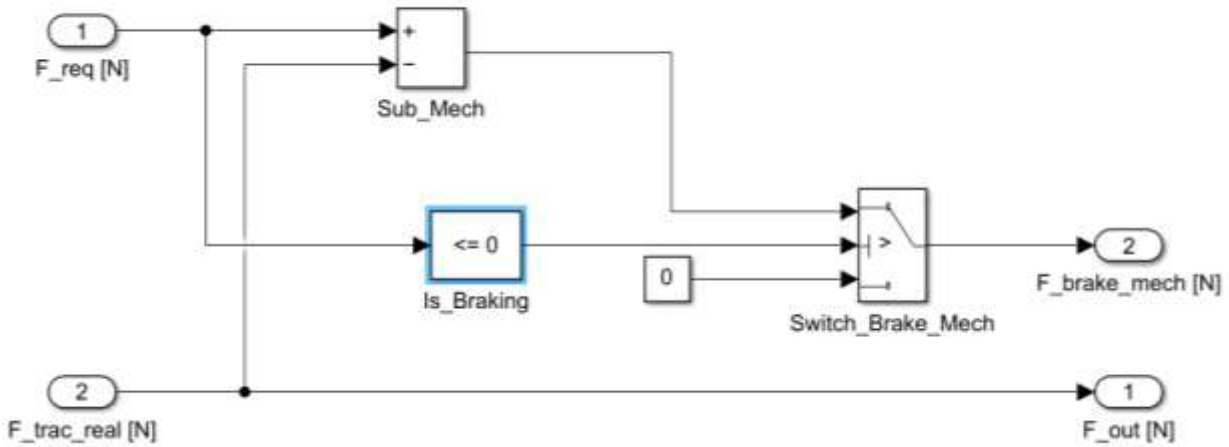


Gear shifter



Feature / Metric	E-Mode (Pure Electric)	ICE / Hybrid / Race Modes
Primary Objective	Maximize electric range & efficiency (Wh/km); quiet urban cruising	Keep ICE within high-efficiency / high-boost powerband
1st Gear Function	Short launch gear (0 -> 18 km/h) solely to start motion from standstill	Full acceleration gear to build boost pressure
Shift Thresholds [km/h]	**1st: 0-18 | 2nd: 18-79 | 3rd: 79-126 | 4th: >126**	**1st: 0-65 | 2nd: 65-119 | 3rd: 119-173 | 4th: >173**
Disconnect Clutch	OPEN (ICE decoupled at 0 RPM, zero drag / pumping losses)	CLOSED (ICE mechanically coupled to EM & gearbox input shaft)
Physical Rationale	Electric machine produces peak torque from 0 RPM. Shifting to 2nd early avoids high motor RPM, drastically reducing iron losses & back-EMF	ICE produces minimal low-RPM torque and suffers turbo lag. 1st gear must be held to ensure the upshift lands above 2500 RPM.

Brake Blending



Control Formulation:

- **Is_Braking Check:** If F_req > 0 (traction), F_brake_mech = 0 N.
- **Residual Subtraction:** F_brake_mech = F_req - F_trac_real. Both signals are negative in braking, so |F_mech| = |F_req| - |F_regen|.

Why This Architecture?

1. Maximum Energy Harvesting ("Regen-First")

The electric motor is prioritized as the primary deceleration source up to its continuous limit. Captures >80% of urban braking energy directly into the battery pack, this eliminates pad/disc friction wear and toxic particulate emissions, keeping brake discs cool and fade-free for high-demand track braking zones.

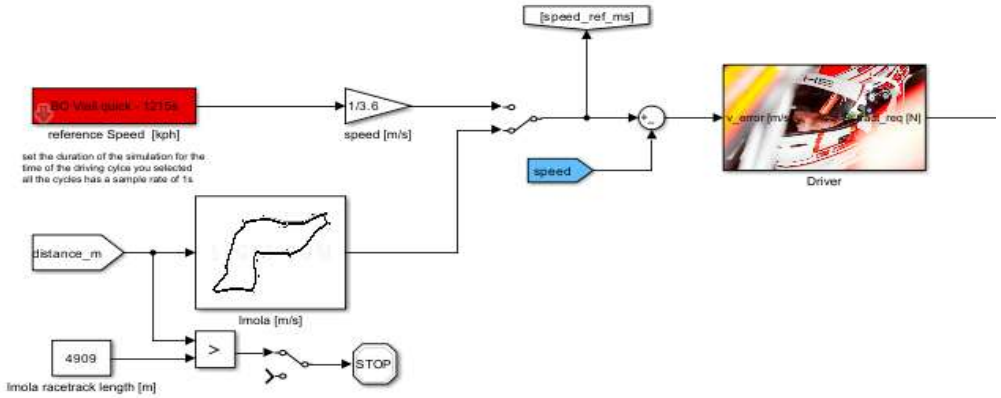
2. Residual Mechanical Actuation

Mechanical calipers only intervene to provide the residual shortfall: F_mech = F_req - F_trac_real. Friction pads engage only when driver deceleration demand exceeds electric machine capacity.

3. Seamless Driver Pedal Feel

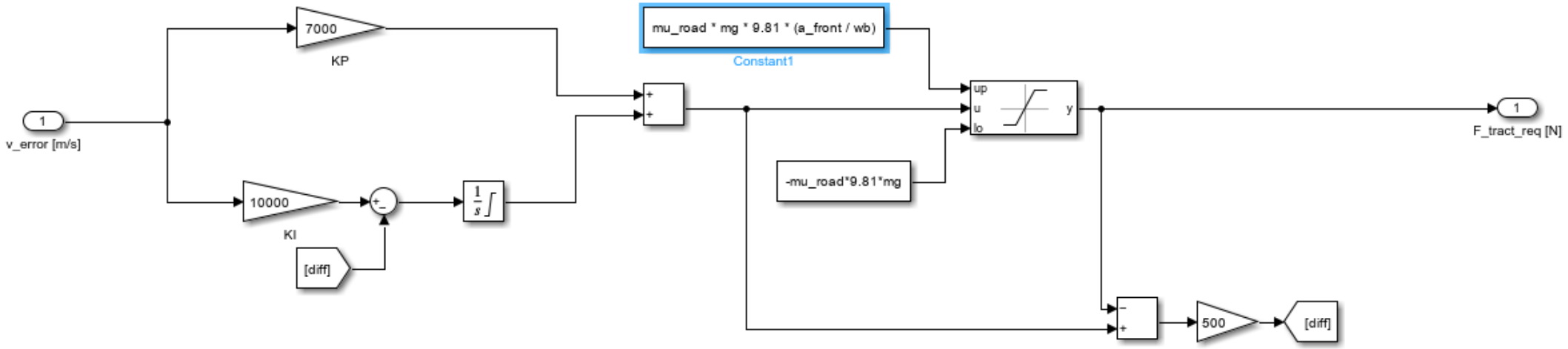
Continuous residual subtraction guarantees that total braking torque strictly matches driver intent (F_tot ≡ F_req).

Driver Model



PI Controlled Traction request generator:

The driver model is a PI controller driven traction force request generator. The error between the reference and the actual speed is used as the input to the PI regulator. The maximum traction force for braking and ting conditions is limited by the maximum available grip.



Lap Time Simulator

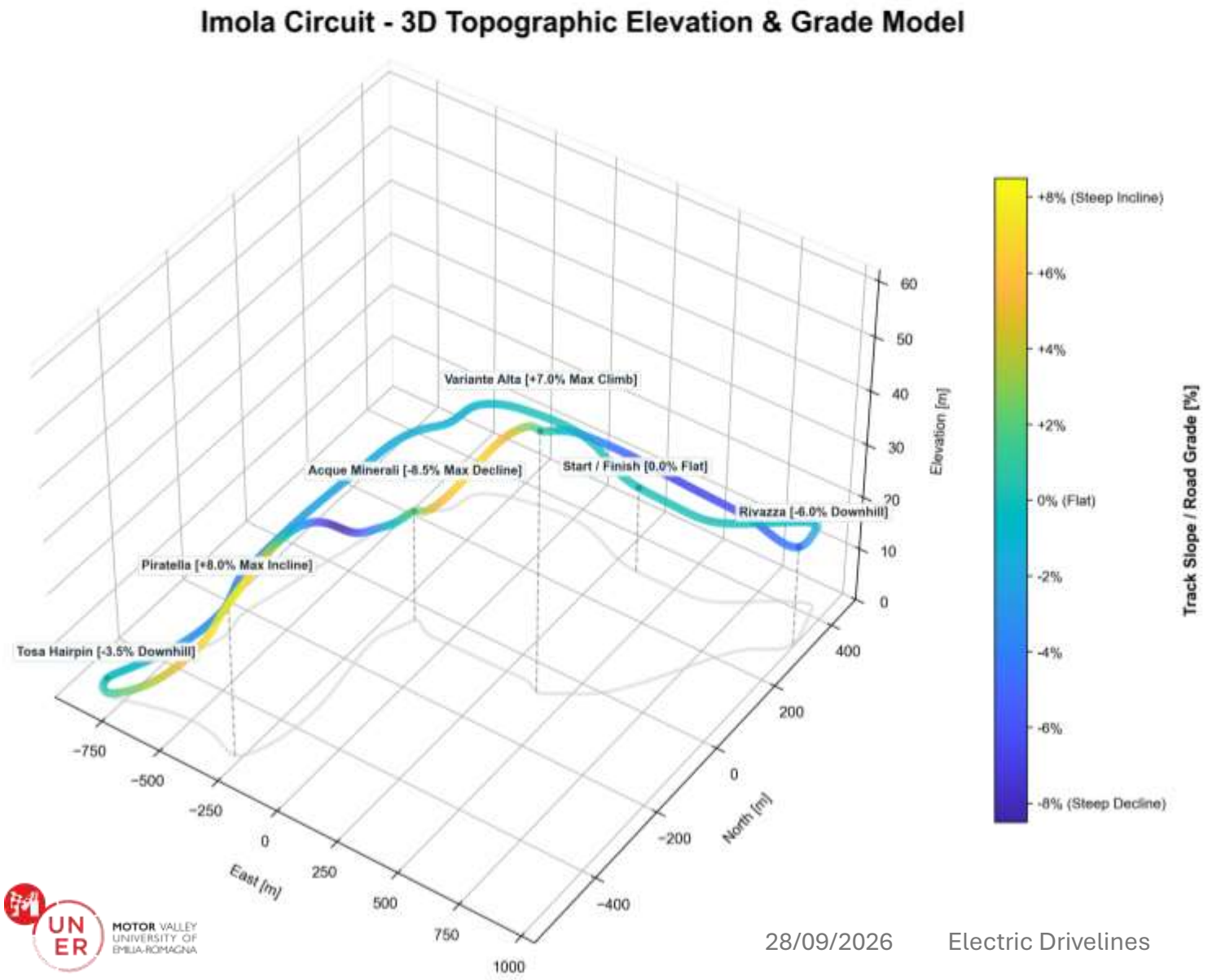


- Personal lap time simulator used to generate speed profiles for Imola Track

```
%% IMOLA Racetrack Speed Reference
s_pts = [0, 350, 450, 850, 950, 1350,
1450, 2000, 2400, 2500, 3000, 3100,
3800, 3900, 4100, 4600, 4700, 4909];

v_kph = [170, 220, 125, 195, 130,
175, 85, 135, 185, 105, 180, 110,
205, 100, 115, 195, 90, 170];
```

Lap Time Simulator



- Racetrack trajectory, vehicle speed and track inclination angle allow to simulate the real track conditions

```
%% IMOLA Racetrack Grade
grade_pts = [0, 0, 0, 0,
0.0150, 0.0300, 0.0500,
0.0450, -0.0300, -0.0908,
0.0935, 0.0200, -0.0400, -
0.0350, -0.0200, -0.0100, 0,
0]; % pu
```

Electric Motor - Selection

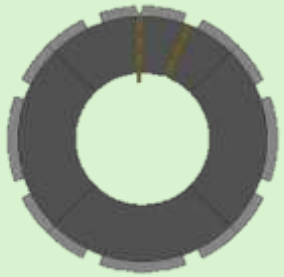
Needs:

• High power density

- Strict axial packaging envelope in the Porsche 930 rear overhang (< 100 mm bellhousing gap).

- Axial-flux disc architecture delivers > 5.4 kW/kg specific power density.

- Extremely low mass (~12.5 kg) preserves the historic 40/60 rear-biased vehicle dynamics.



• Linear torque curve

- Instantaneous 230 Nm plateau available from 0 RPM via pure FOC ($i_d = 0$).

- Directly infills the 0.8 s spool-up lag crater of the single massive KKK K27 turbocharger.

- Isotropic air gap ($L_d = L_q$) yields zero reluctance cross-coupling and linear driver throttle response.



• Peak power at ICE high rpm

- Maintains full overboost power (67.5 kW) up to the 6700 RPM Porsche 930 engine redline.

- Low isotropic inductance prevents high-speed armature reaction from shrinking voltage limit circles.

- Seamless handover to pressurized Flat-6 ICE without premature high-RPM power collapse.



Electric Motor - Selection

SAME Volume SAME Supply Comparison:

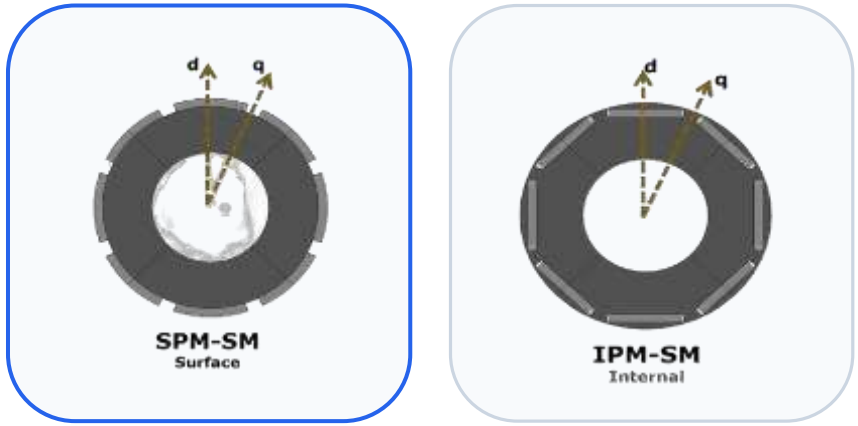
	SPM-SM	IPM-SM	VIPM-SM	RIPM-SM	PMAR-PM
inductance Ld [mH]	0,82	1,03	1,10	0,58	0,50
inductance Lq [mH]	0,84	1,69	2,09	0,89	2,09
flux linkage by magnets [V·s]	0,18	0,15	0,12	0,22	0,08
magnet mass [kg]	2,60	2,60	2,50	4,20	3,00
a=fluxM/(Ld*In)	1,83	1,21	0,90	3,16	1,33
rated torque [Nm]	259	240	223	321	224
base speed [rpm]	1610	1550	1490	1410	1630
rated power [kW]	50,3	50,4	46,8	50,2	50,4
max. torque (x2In) [Nm]	520	571	605	666	718
max. power (x2In) [kW]	92,0	68,1	60,1	100,1	87,0



Electric Motor - Selection

SAME Volume SAME Supply Comparison:

- Supply voltage: 345.6 V (Nominal 96s2p Battery Pack)
- Max current: 139.4 A (Overboost S2 Peak Limit)

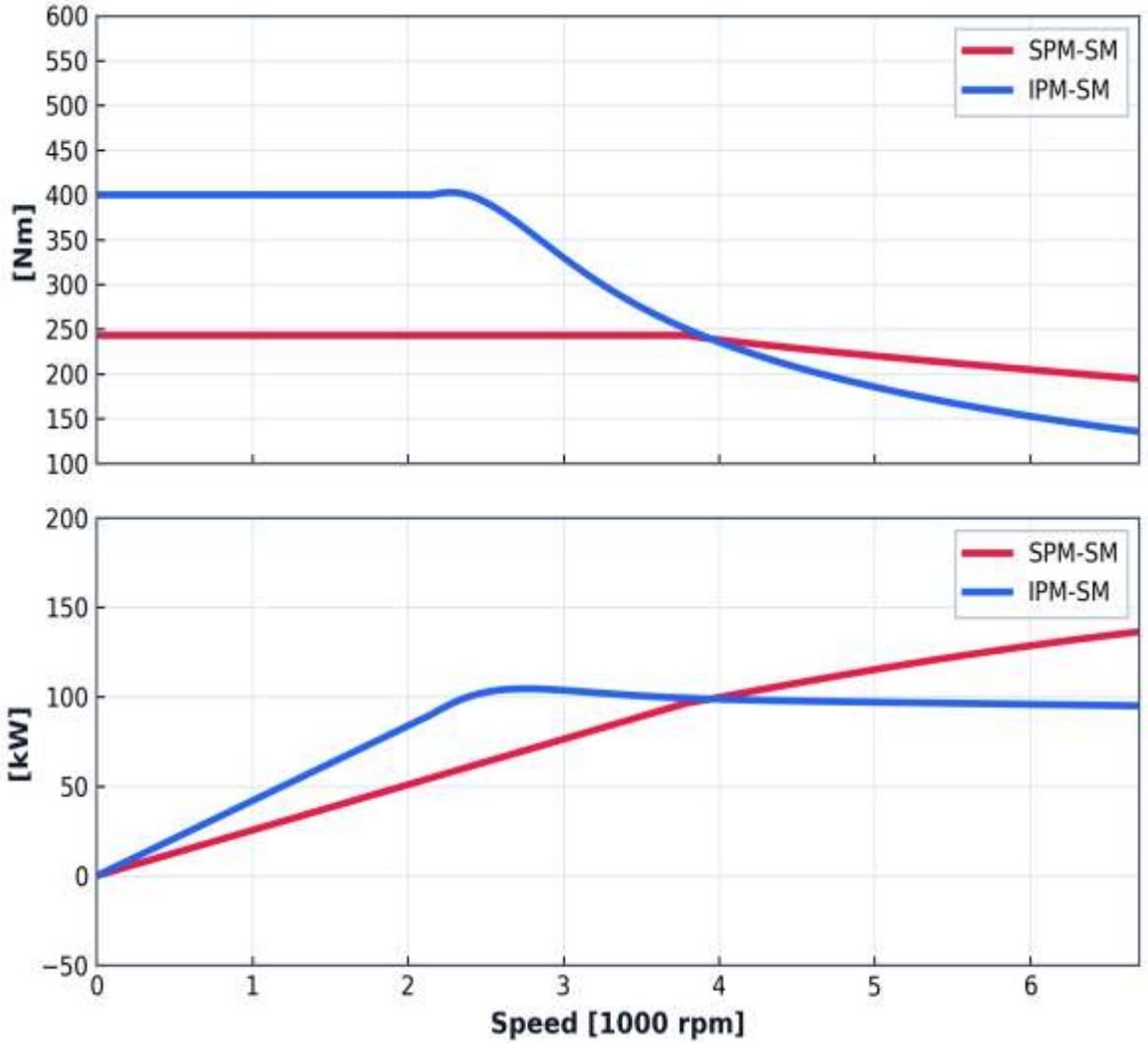


- **SPM-SM Physical Advantage:**

Uniform airgap gives equal reluctance in d and q axes (L_q = L_d). Pure Lorentz torque allows constant power to extend smoothly to high RPM without deep demagnetizing currents.

- **IPM-SM Limitation:**

Iron paths around magnets create high q-axis inductance ($L_q > L_d$). Reluctance torque aids low speeds, but high $\omega L_q i_q$ voltage drop forces severe field weakening and early power collapse.



Electric Motor - Selection

Motor Parameters

Motor data	Value	Unit
Source voltage	345.6	V
Rated current	85	A
Rated torque	150	Nm
Rated power	50	kW
Max speed	6700	rpm
Overload factor	1.64	-
Max current	139.4	A
Max torque	230	Nm
Max power	67.5	kW
L_d	0.5	mH
r	1	-
Pole pairs	6	-
Flux coefficient (a)	1.5	-

TECHNICAL SPECIFICATIONS & SIZING RATIONALE

1. Direct Redline Matching (n_{MAX} = 6700 rpm):

Motor maximum speed matches the air-cooled Flat-6 crankshaft redline directly (1:1 P2 coupling), completely avoiding the weight, packaging penalty, and mechanical friction losses of an intermediate reduction gearbox.

2. Dual Operating Regimes (S1 Continuous vs. S2 Overboost):

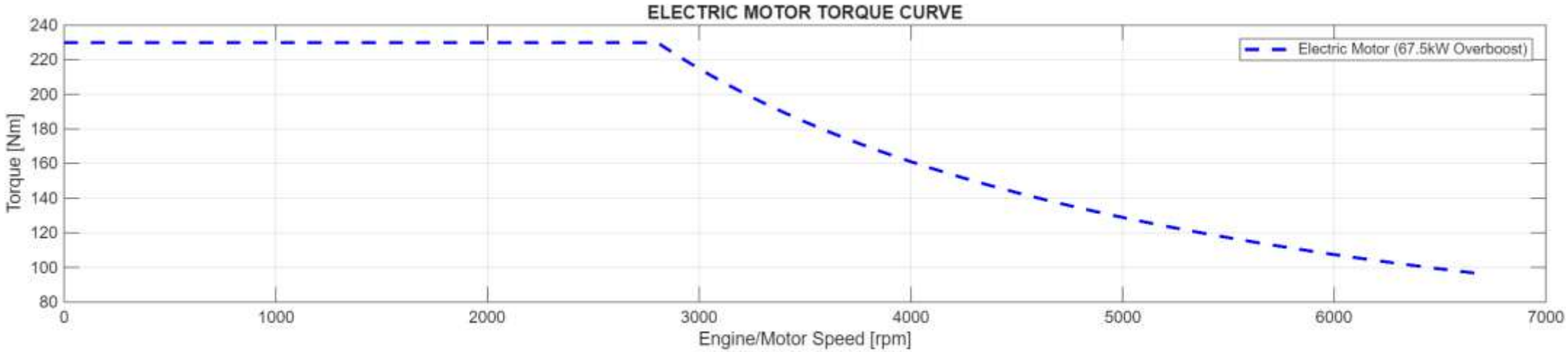
- Continuous S1 (50 kW / 150 Nm @ 85 Arms): Thermally sustainable indefinitely for sustained cruising and regenerative braking.
- Overboost S2 (67.5 kW / 230 Nm @ 139.4 Arms): Overload factor 1.64x active during full throttle acceleration.

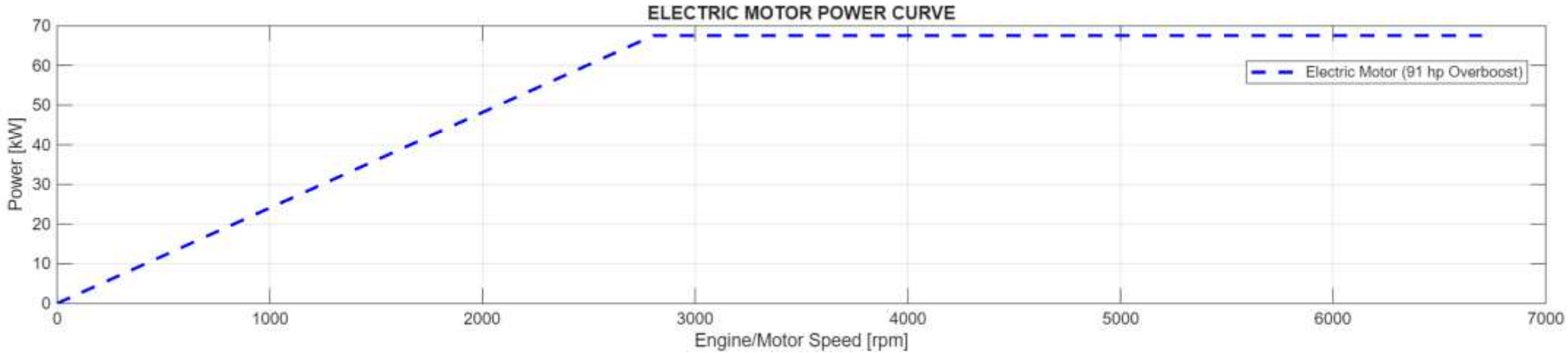
3. FOC Vector Control Simplicity (r = 1.0):

Because saliency ratio r = L_q / L_d = 1.0, reluctance torque is strictly zero. Optimal MTPA trajectory reduces to pure quadrature control (i_d = 0, i_q = I_0), minimizing inverter switching losses and processing overhead.

Electric Motor - Selection

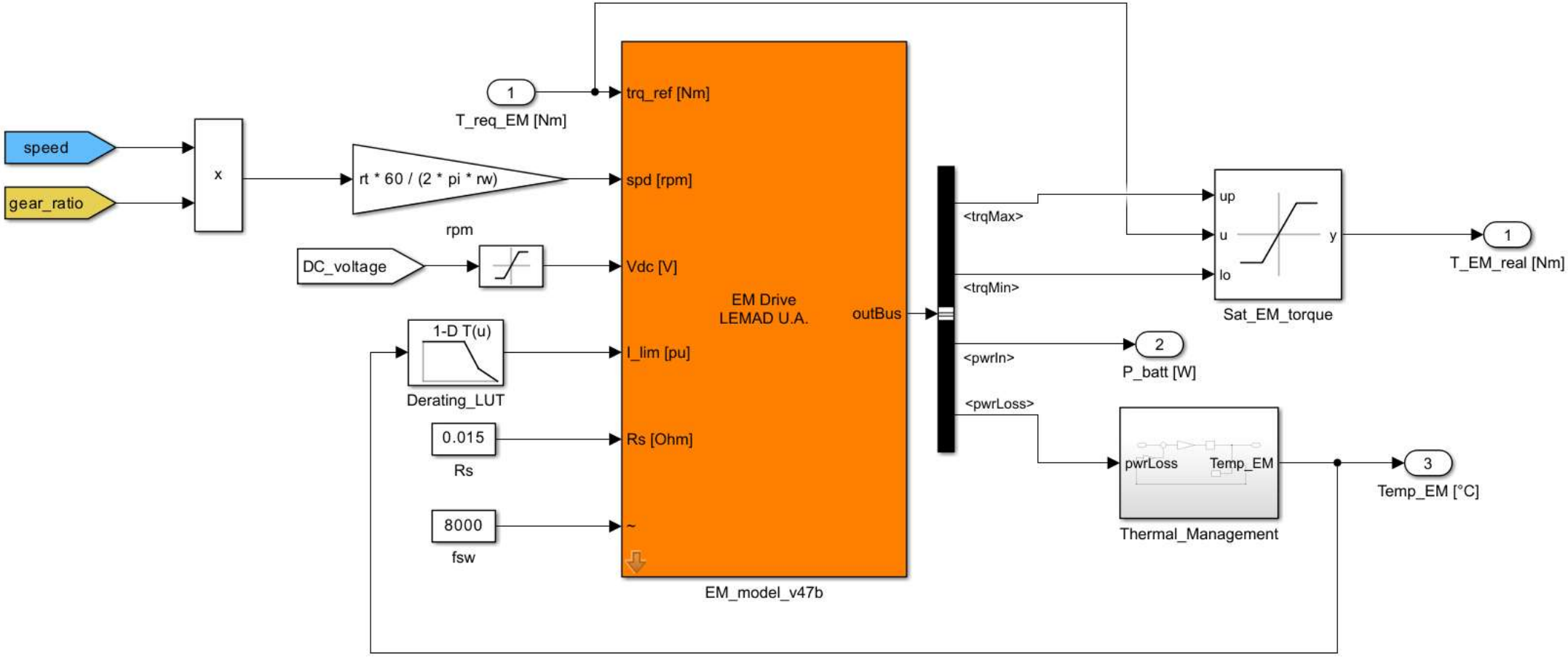
Motor torque and power curve





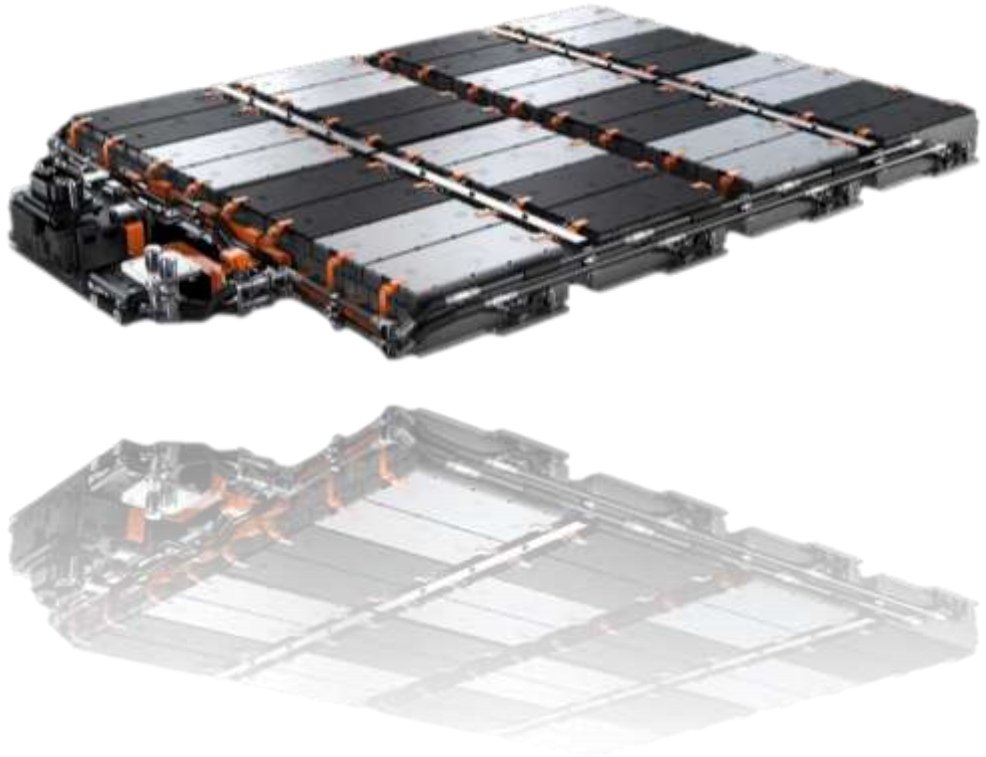
Electric Motor - Modelling

Simulink Scheme:



Battery Pack - Requirements

To eliminate the Porsche 930's turbo lag with instant torque while respecting the strict 26 kg weight budget in the front trunk (frunk), cell selection is governed by five key project requirements:



1. Extreme Continuous C-Rate (≥ 10C / ≥ 40 A per cell)

Required to supply 67.5 kW peak overboost (230 Nm) from a compact 2.90 kWh pack (96S2P). Demands up to 37.25 A/cell peak on corner exits at Imola.

2. Ultra-Low Internal Resistance (R0 ≤ 10 mΩ)

Essential for our 100% Air-Cooled design. Drastically cuts Joule heating ($I^2 \cdot R0$), keeping cell temperatures under 30°C on track without heavy liquid cooling.

3. High Gravimetric Power Density (≥ 2.2 kW/kg)

Strict 26 kg pack mass budget (13.4 kg cells) in the front trunk (Frunk). Rebalances the chassis towards an optimal 41.5% F / 58.5% R weight distribution.

4. High Fast-Charge Acceptance for Regenerative Braking

Must absorb intense deceleration pulses (up to -3500 N / 28.6 kW regen) during heavy braking (e.g: into Tosa and Rivazza) without cell overvoltage.

5. Cylindrical Form Factor (Passive Safety & Airflow Channels)

Rigid steel can provide impact protection in the front crush zone. Natural inter-cell gaps create convective airflow channels for aerodynamic frunk air cooling.

Battery Pack – Cell Definition & Characteristics

Cell selected: *Molicel INR-21700-P42A* (High-Power Cylindrical Li-Ion Cell) [NMC]

Electrical Parameters	Value [unit]
Nominal Capacity	4200 [mAh] (4.2 Ah)
Nominal Voltage	3.6 [V]
Nominal Energy	15.12 [Wh]
Internal Resistance (1kHz)	≤ 15 [mΩ] (DC: 45 mΩ)
Voltage Range	2.50 – 4.20 [V]
Max Continuous Discharge	45 [A] (10.7 C)
Peak Discharge Current (Pulse)	100 [A] (24 C)
Standard Charge Current	8.4 [A] (2 C)

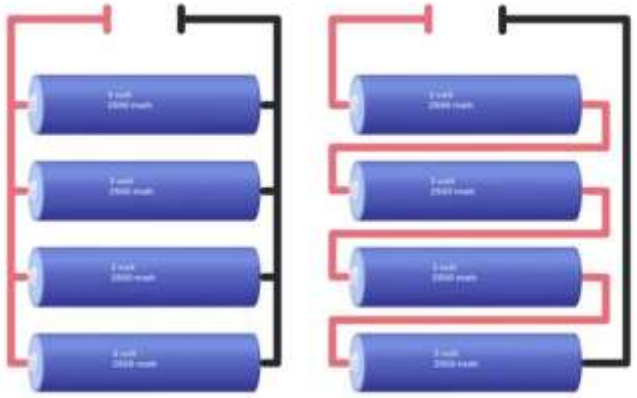
Mechanical Parameters	Value [unit]
Mass	0.067 [kg]
Diameter	21.4 [mm]
Height	70.0 [mm]
Format	Cylindrical 21700

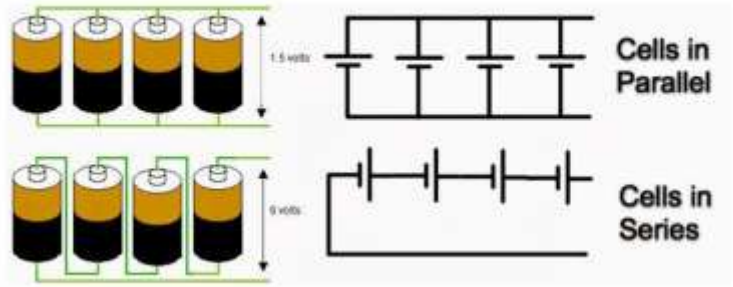


Selection Rationale:

- Highest continuous discharge rate (45A) in the 21700 class
- Outstanding thermal stability under repetitive overboost pulses
- Low internal impedance (45 mΩ) minimises Joule heating
- High volumetric power density for front-trunk packaging

Battery Pack – Dimensioning





- **Number of cells in series:**

$$N_{SERIES} = \frac{V_{DC}}{V_{NOM_CELL}} = \frac{345.6\ [V]}{3.6\ [V]} = 96$$

- **Total number of cells:**

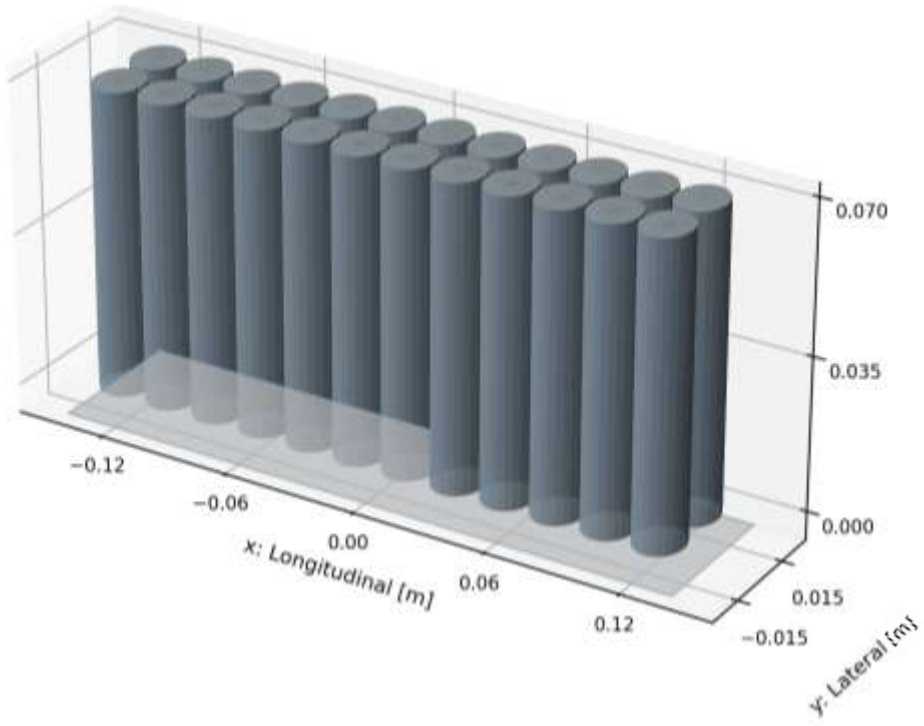
$$N_{TOTAL} = \frac{P_{PEAK}}{V_{NOM_CELL} \cdot I_{DISCHARGE}} = \frac{67.5\ [kW]}{3.6\ [V] \cdot 100\ [A]} \approx 192$$

- **Number of cells in parallel:**

$$N_{PARALLEL} = \frac{N_{TOTAL}}{N_{SERIES}} = \frac{192}{96} = 2$$

Resulting Pack Configuration: 96S 2P (192 cells total)
*Continuous electrical capability: 192 * 3.6V * 45A = 31.1 kW*
*Over boost pulse capability: 192 * 3.6V * 100A = 69 kW*

Battery Pack – Module Sizing



Considering the volume available in the Porsche 930 front luggage compartment (Frunk) the optimal architecture divides the 96S pack into 8 modules in series:

$$N_{MODULES} = 8$$

$$N_{SERIES_MODULES} = \frac{N_{SERIES}}{8} = \frac{96}{8} = 12$$

Each module will be 12s2p (24 cells per module) all connected in series.

This will result in a nominal voltage per module of:

$$V_{NOM_MODULES} = 12 * 3.6 = 43.2\ [V]$$

Which is admissible, touch-safe value (< 60 V DC limit according to ISO 6469 / ECE R100).

Battery Pack – Dimensions

Considering the scheme chosen for the modules (12s2p) and the physical dimensions of each 21700 cell:

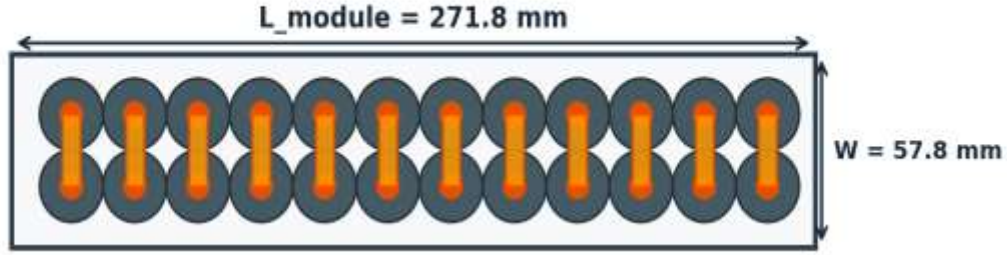
- **Diameter = 21.4 mm**
- **Height = 70.0 mm**

We calculate the physical dimensions of each module, adding 15 mm to consider the external enclosure, structural cell spacers, and busbar covers:

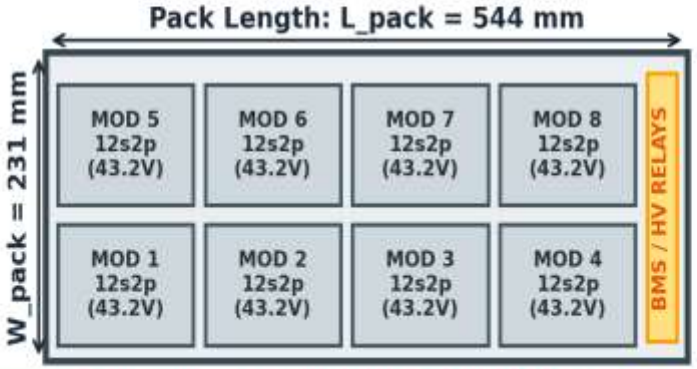
- **L_module = 21.4 * 12 + 15 = 271.8 mm**
- **W_module = 21.4 * 2 + 15 = 57.8 mm**
- **h_module = 70.0 + 15 = 85.0 mm**

Packaging Layout in Frunk:
Arranged in 2 rows of 4 modules, yielding an ultra-compact pack footprint of only 544 mm (L) x 231 mm (W) x 85 mm (H),
fitting into the 930 front luggage compartment and rebalancing chassis weight distribution from 39/61 to 41.5/58.5%.

Single Module (12s2p) - Mechanical Outline & Busbars



Porsche 930 Complete Pack Layout (8 Modules in Series = 96S 2P)



Front Luggage Compartment (Frunk) Installation | Low CoG | Weight: 26.0 kg

Battery Pack – Final Electrical Quantities

Each module will be characterized by a number of cells of:

$$N_{CELLSxMODULE} = N_{SERIES_MODULE} \cdot N_{PARALLEL_MODULE} = 12 \cdot 2 = 24$$

And a total mass per module (including enclosure & busbars):

$$m_{MODULE} = N_{CELLSxMODULE} \cdot m_{CELL} + m_{CASE} = 24 \cdot 0.067 + 0.89 = 2.5\ [kg]$$

Which results in a complete battery pack having:

$$V_{BATTERY_NOM} = N_{MOD} \cdot V_{NOM} * N_{SERIES_{MODULE}} = 8 \cdot 3.6 \cdot 12 = 345.6\ [V]$$

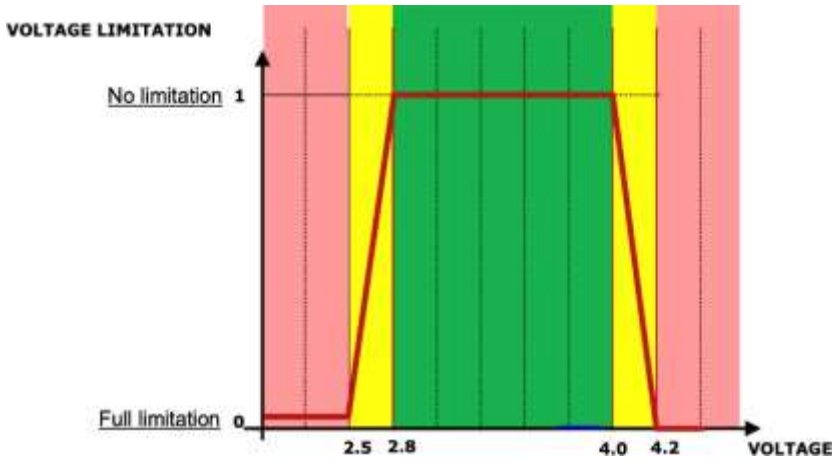
$$P_{BATTERY_PEAK} = N_{TOT} \cdot V_{NOM} * I_{DISCHARGE} = 192 \cdot 3.6 \cdot 97.6 = 67.5\ [kW]$$

$$E_{BATTERY} = N_{SERIES} \cdot V_{NOM} \cdot C_{NOM} \cdot N_{PARALLEL} = 96 \cdot 3.6 \cdot 4.2 \cdot 2 = 2.90\ [kWh]$$

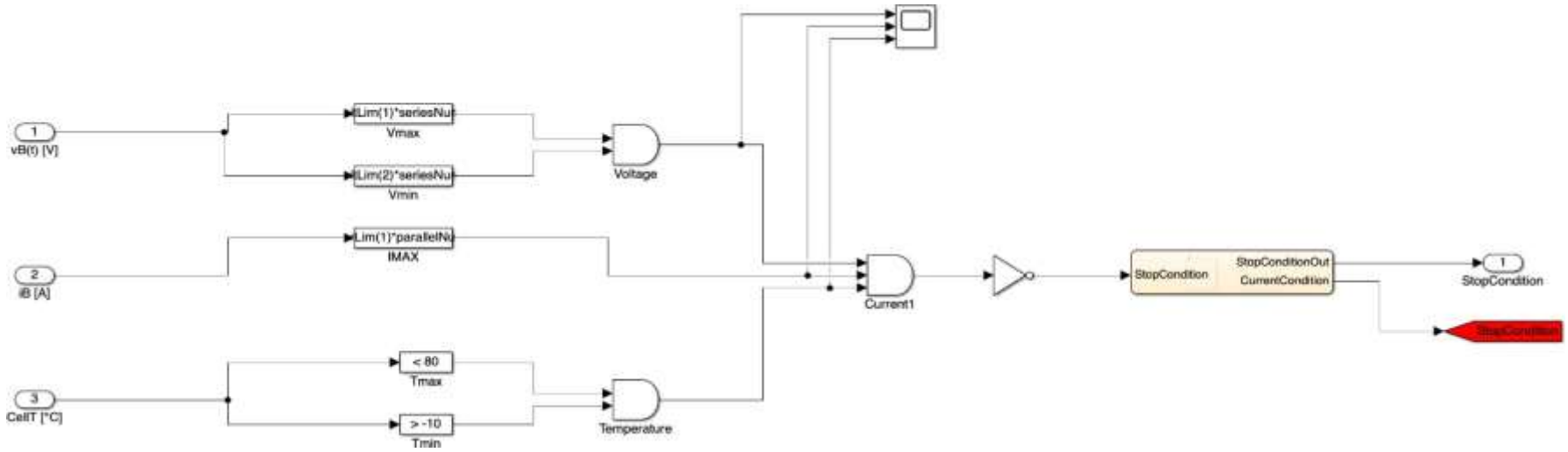
$$m_{BATTERY} = 26\ [kg]$$ (12.9 kg cells + 13.1 kg packaging, BMS & active cooling)

Configuration	Nominal Voltage	Peak Power	Gross Energy	Total Pack Mass	Frunk Footprint
96S 2P (8 Mod)	345.6 V	67.5 kW (Overboost)	2.90 kWh	26.0 kg	544 x 231 x 85 mm

Battery Pack – BMS

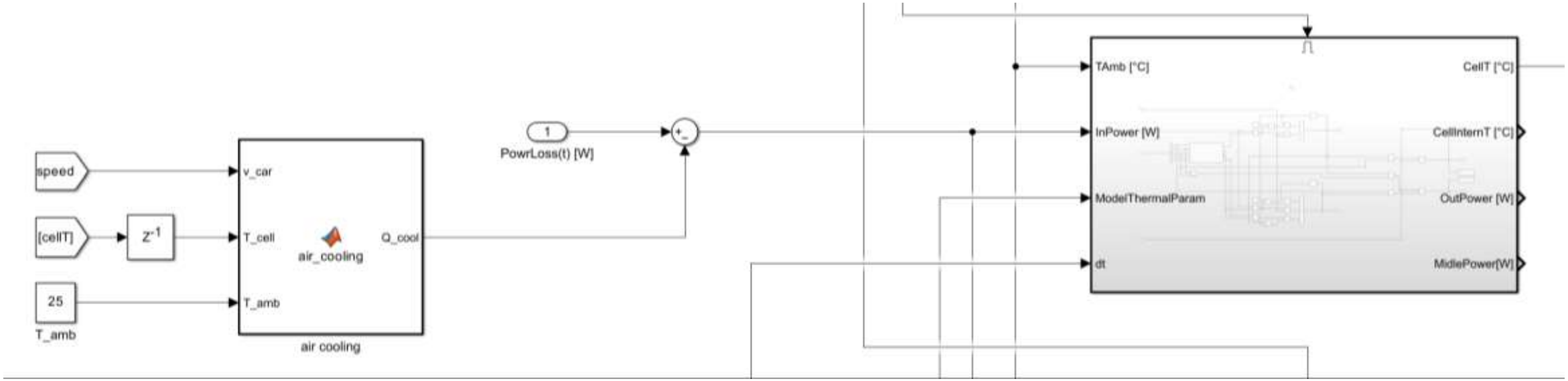


To ensure the proper functioning of the battery and improving its **safety** a BMS is implemented as a **STOP signal** in case the voltage, current or thermal limits are exceeded.



Battery Pack – Cooling System

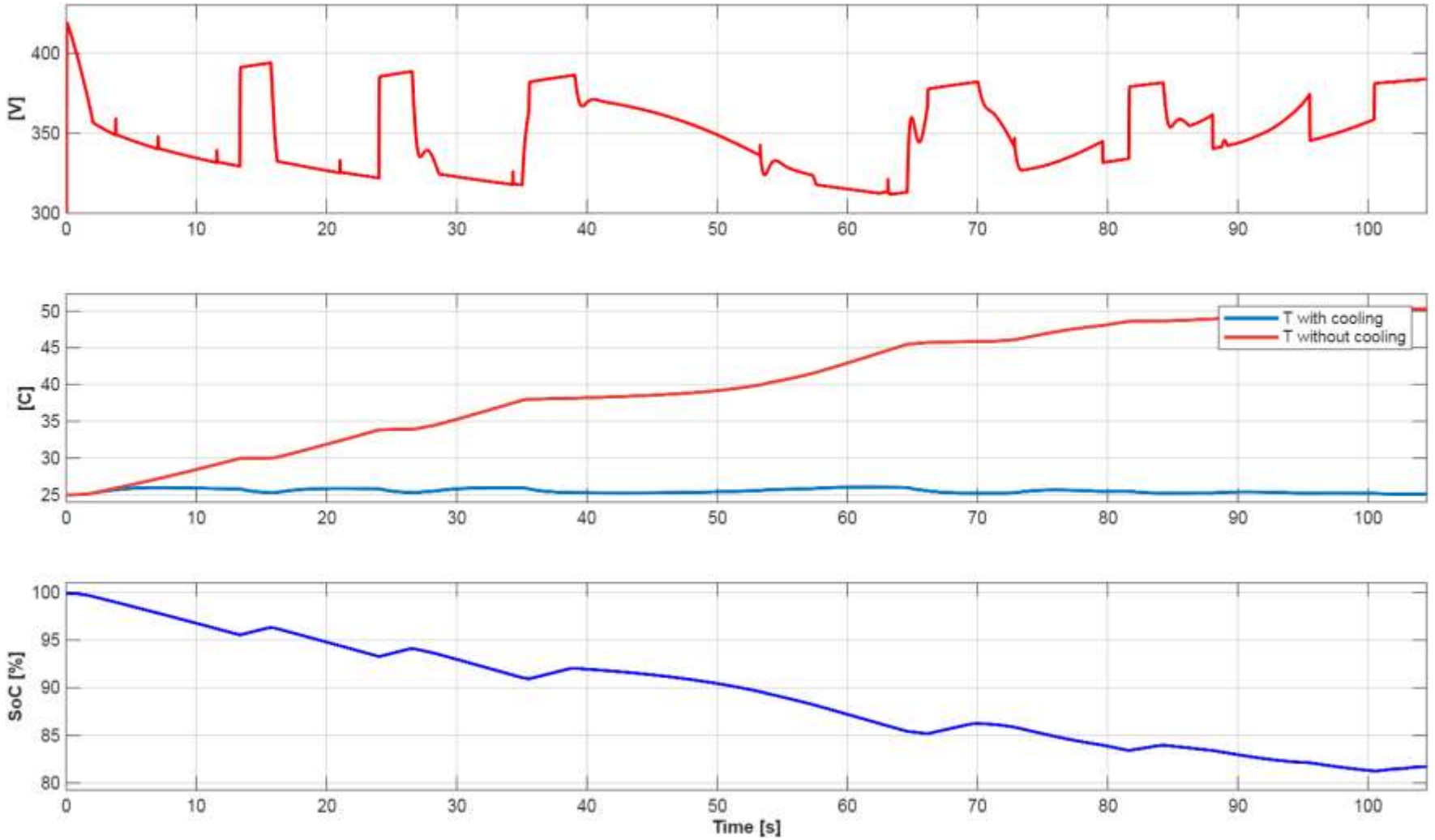
Automotive batteries can be cooled mainly in two different ways; **Air Cooling** or **Water Cooling**, considering the focus on **performance**, the high speeds reached on track and the need to not increase too much the weight, **air cooling** seems to be the best preliminary choice.



A simple **air-cooling** is modelled following the formula: $P_{COOLING} = h \cdot A \cdot (T_{BATT} - T_{AIR})$

Battery Pack – Cooling System

Thermal analysis of the battery is carried out:



The plot shows one push lap around Imola with and without cooling. It is evident that:

- The **battery needs some cooling;**
- The implemented cooling works and it is enough for our application

Battery Pack – Cost and Weight

TOTAL SYSTEM MASS

As seen before the estimated weight includes cells, packaging, BMS, HV safety & air cooling:

$$m_{BATTERY} = 26\ [kg]$$

Component / Subsystem	Hardware Specs	Mass	%
Active Cells (Molicel P42A)	192x 21700 (96S2P, 45A CDR)	**13.44 kg**	**51.7%**
BMS & Sensing Harness	96S master/slave, 96 taps, 16 NTC	**1.80 kg**	**6.9%**
HV Safety Box & BDU	Gigavac contactors, MSD, fuse	**2.50 kg**	**9.6%**
Structural Case & Busbars	Al 6061 tray, carbon lid, Cu bars	**6.76 kg**	**26.0%**
Dedicated Air-Cooling	Dual 12V blowers, ducts, pads	**1.50 kg**	**5.8%**
Total Battery Pack Mass		**26.00 kg**	



TOTAL PACK MASS
26.0 kg
51.7% Cell Efficiency

TOTAL SYSTEM COST
€2,050
€706 / kWh gross

AIR-COOLING SAVINGS
-5.0 kg & -€335
vs Liquid Cooling

ENERGY CAPACITY
2.90 kWh
96S2P • 345.6 V

TOTAL BILL OF MATERIALS

Cell retail is ~8.50 €, but drops to 4.20 € for small-batch OEM orders

Component / Subsystem	Hardware / Supply Basis	Cost	%
Active Cells (192x P42A)	Tier-1 21700 cells (192 x €4.20 OEM)	**€806.40**	**39.3%**
Automotive 96S2P BMS	Master/slave board, current shunt, CAN	**€450.00**	**21.9%**
HV Safety Box	Dual contactors, precharge relay	**€360.00**	**17.5%**
Enclosure & Busbars	Tray, carbon lid, Cu bars	**€290.00**	**14.1%**
Air Cooling Hardware	Dual 12V blowers, bumper ducts, pads	**€145.00**	**7.1%**
Total System BOM Cost		**€2,051.40**	

Hybridization – Cost and Weight

Weight Assessment

P2 Coaxial Architecture vs. Generic Benchmark

In order to estimate the weight, approximations based on our P2 architecture are made:

- Motor mass (SPM-SM): 12.5 kg;
- Dog / Clutch mass (P2 K0 disconnect): 6.0 kg;
- Coupler mass (Coaxial 1:1 adapter): 3.5 kg;
- Wiring, power electronics: 11.5 kg;
- Cooling system mass (Air-cooled): 3.5 kg.
- E-Reverse → No reverse gear: -4.0 kg.

Which, together with the battery (26.0 kg), results in a total estimated weight of 59.0 kg.

Cost Assessment

Prototype Bill of Materials (BOM) & Sourcing

In order to estimate the cost, approximations based on prototype BOM are made:

- Motor cost (SPM-SM): € 3,200;
- Dog / Clutch cost (P2 K0 disconnect): € 650;
- Coupler cost (Coaxial 1:1 adapter): € 450;
- Wiring, power electronics, safety cost: € 3,750;
- Cooling system cost (Air-cooled): € 240.

Which, together with the battery (€ 2,050), results in a total estimated cost of € 10,340.

ECU - Driving Modes



E-MODE

K0 Clutch: OPEN (ICE Off)
- ICE: 0 kW (Disconnected)
- EM: Up to 230 Nm / 50 kW

CONTROL STRATEGY
- 100% Electric if SoC > 15%
- No ICE drag or pumping losses
- Emode gear shifts

BRAKE REGENERATION
Full Regen Active

Urban & ZTL Commuting



ICE-MODE

K0 Clutch: CLOSED
- ICE: 0 kW (Disconnected)
- EM: Up to 230 Nm / 50 kW

CONTROL STRATEGY
- Conventional 3.3L Turbo
- Traditional manual shift feel
- Fallback if battery depleted

BRAKE REGENERATION
Regen OFF

Old Style Feelings



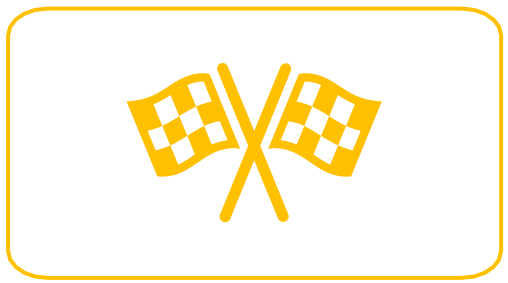
HYBRID-MODE

K0 Clutch: CLOSED
- ICE: Shifted to sweetspot
- EM: Dual Role (Boost / Gen)

CONTROL STRATEGY
- Load point shifting: ICE recharges battery at 150 Nm
- > 380 Nm: EM provides boost
- Voltage Guard: 280 V ≤ V_DC ≤ 395 V

BRAKE REGENERATION
Regen ON & Active SoC Balance

Max WLTP Fuel Efficiency



RACE-MODE

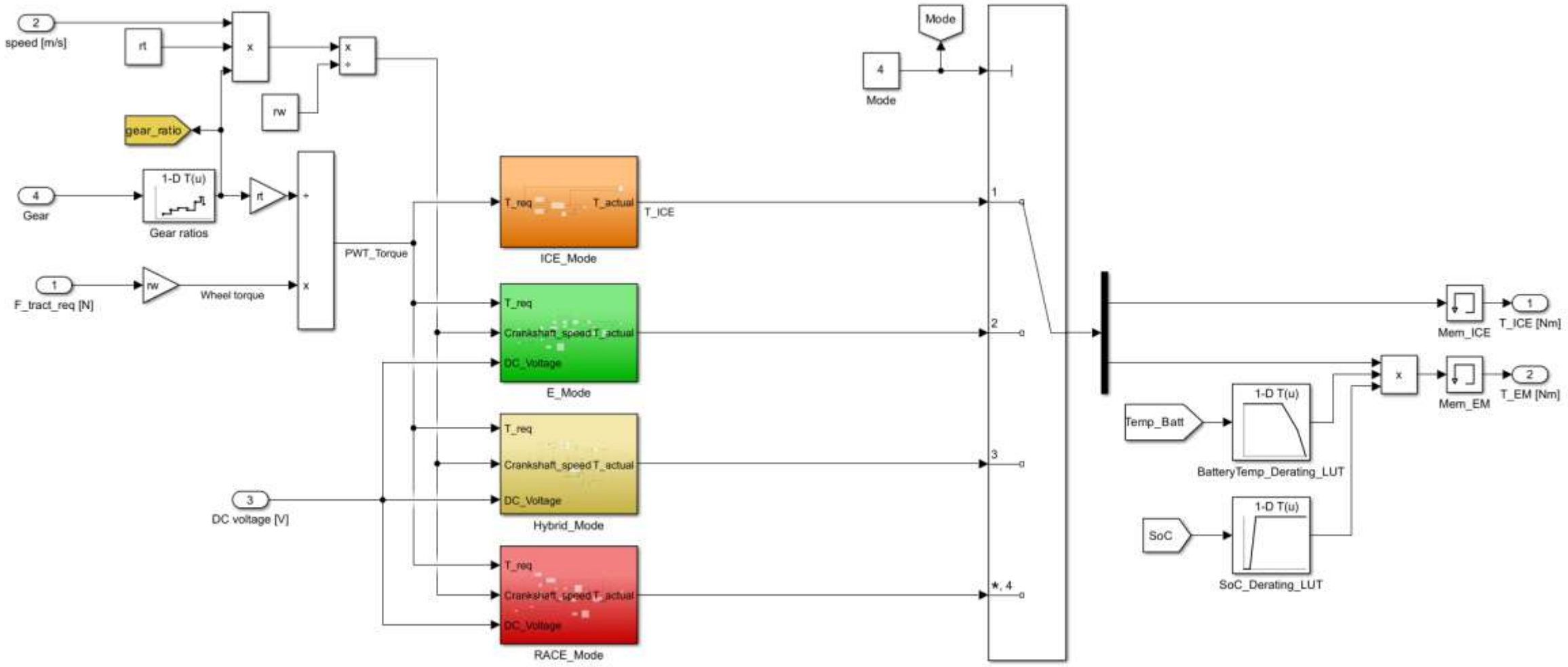
K0 Clutch: CLOSED
- ICE: 100% Throttle (300 hp)
- EM: 67.5 kW Overboost

CONTROL STRATEGY
- Torque Filling (< 2800 RPM): Eliminates single-turbo lag
- Total: 390 hp / 660 Nm combined
- Thermal & SoC derating guards

BRAKE REGENERATION
Regen ON

Lap Time Optimization

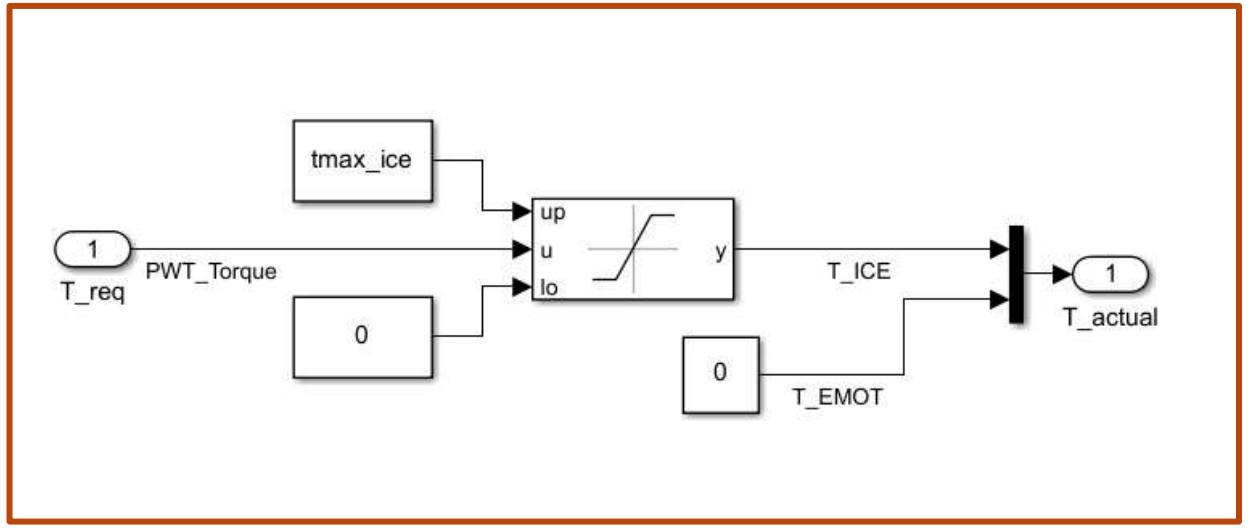
ECU - Driving Modes



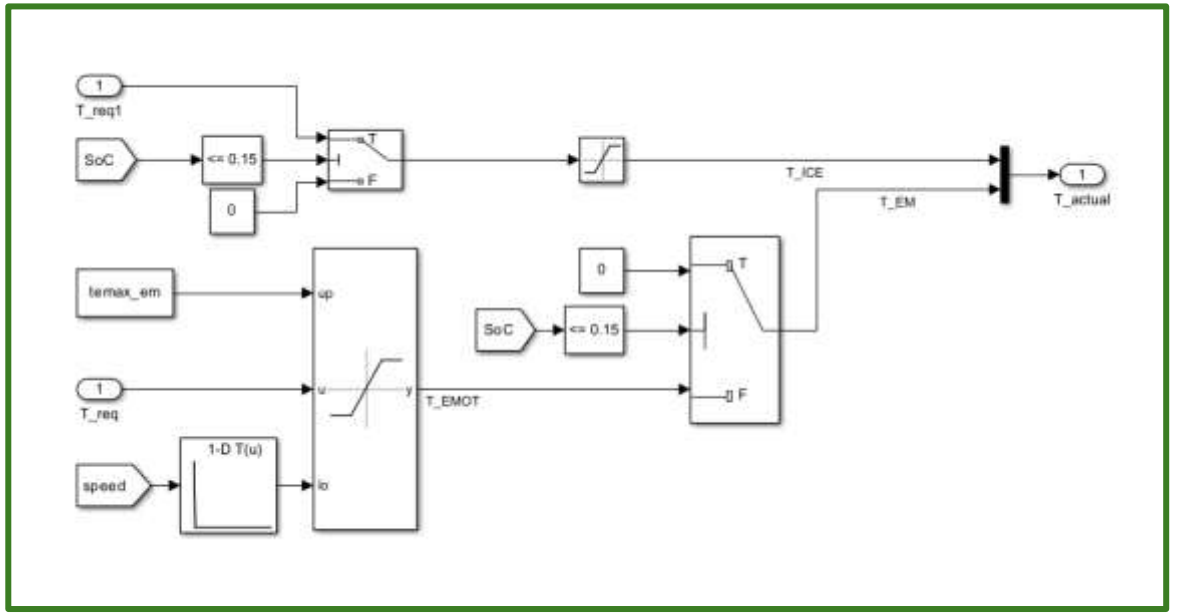
Driving Modes – ICE Mode & EMode

Simulink Schemes:

ICE Mode



EMode



ICE Mode:

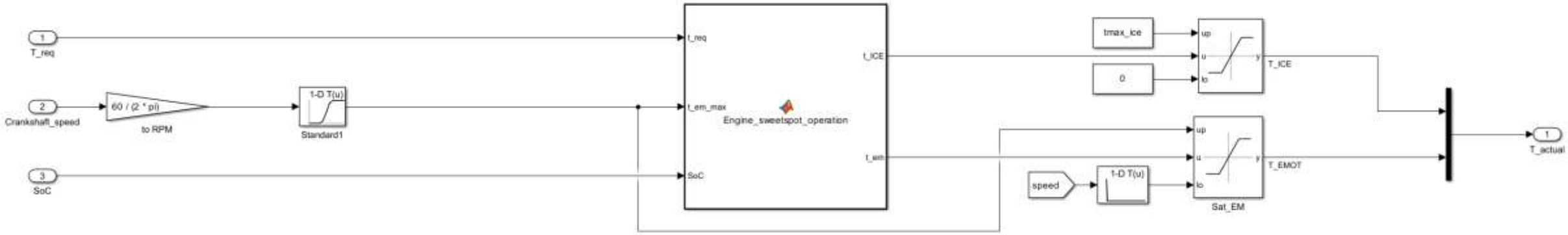
The ICE Mode is the conventional driving of Porsche 930 without any electrical assist. This mode is used for performance comparisons with electric and hybrid modes.

EMode:

EMode provides traction to the wheel solely from the Electric motor and without the engine during normal driving conditions. When the SoC of the battery pack drops below 0.15 %, ICE takes over to match the requested traction force without the Electric motor.

Driving Modes – Hybrid Mode

Simulink Scheme:



Hybrid Mode:

Hybrid Mode combines the ICE and Electric Motor to optimize fuel economy by running the engine at its optimal BSFC point. The Electric Motor covers low-speed driving and delivers torque boost during high acceleration, while absorbing excess engine torque to keep the battery charged.

Driving Modes – Hybrid Mode

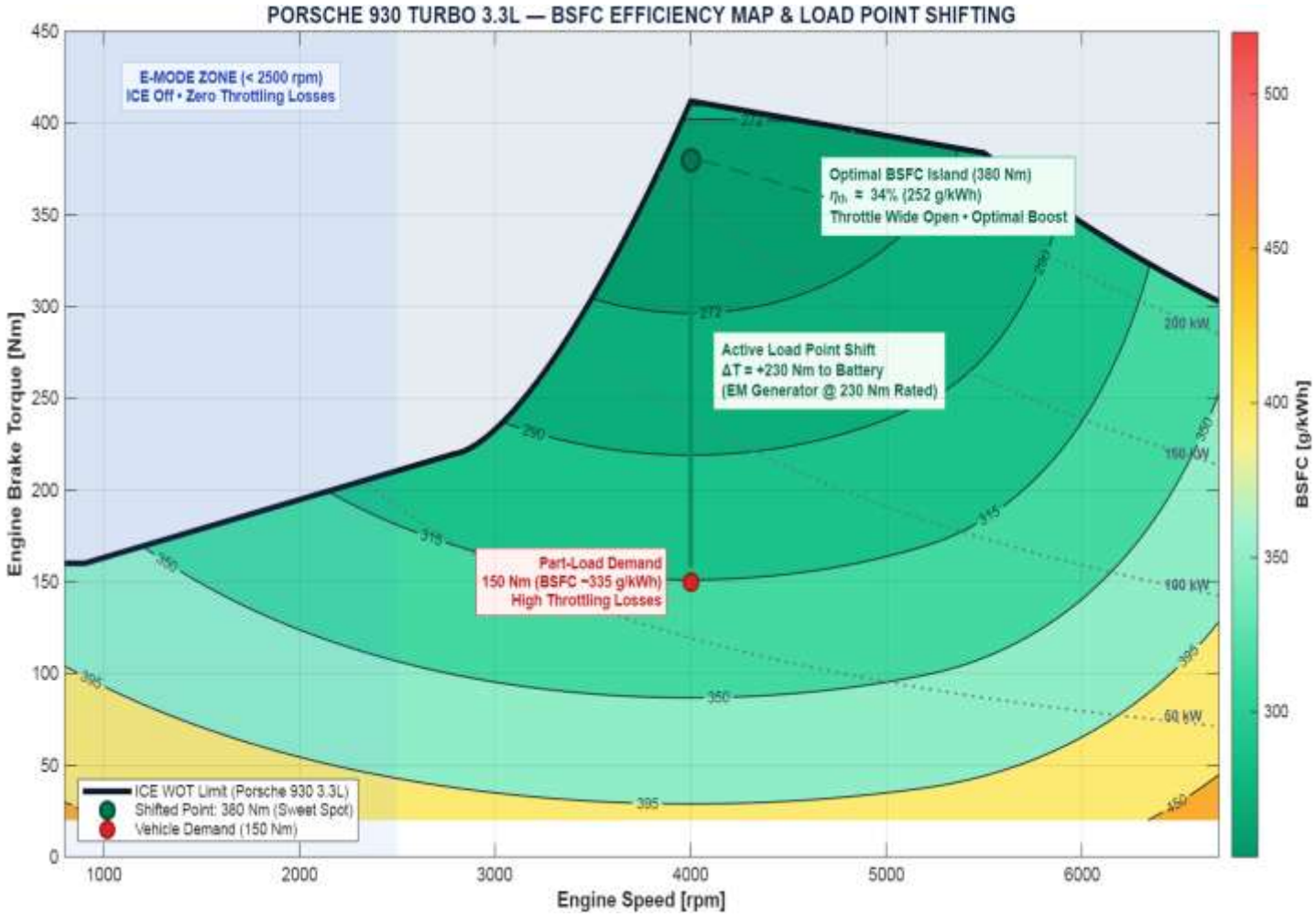
Why does operating at 380 Nm Maximize Efficiency?

Optimal BSFC Island

Internal Combustion Engines reach peak thermal efficiency at specific torque islands (380 Nm). Forcing the engine to run at 380 Nm eliminates low-load throttling losses.

Active load point shifting

When vehicle demand is less than 380 Nm, excess torque is stored as electrical energy in the battery, converting transient engine inefficiency into zero-emission EV driving.



Driving Modes – Hybrid Mode

Hybrid Mode Control Logic:

Control Inputs

- t_req: Requested wheel torque [Nm]
- t_EM_max: Maximum EM torque capacity [Nm]
- SoC: State of Charge of the Battery [%]

Calibrated constants:

- t_sweetspot: 380 [Nm]
- t_lowload_threshold: 110 [Nm]
- SoC_min: 10%

Controlled outputs:

- t_ICE: Requested wheel torque from ICE [Nm]
- t_EM: Requested wheel torque from EM [Nm]

Driving Modes – Hybrid Mode

State Machine Architecture

SoC < 20%

Low-Voltage Recovery:

Forces engine to run at 380 Nm sweetspot to drive generator charging whenever possible until 50% SoC is reached.

SoC> 95%

Overvoltage Guard:

Disables active generator charging to prevent overcharging battery cells.

20% < SoC < 95%

Normal Hybrid:

Standard energy management optimizing fuel economy via pure EV and sweetspot charge sustaining.

Driving Modes – Hybrid Mode

Torque Split across operational regimes:

Torque Request Range	Low - V	Over - V	Normal
t_req < 110 Nm	t_ICE = 380, t_em = t_req - 380	t_ICE = 0, t_em = t_req	t_ICE = 0, t_em = t_req
110 <= t_req <= 380 Nm	t_ICE = 380, t_em = t_req - 380	t_ICE = t_req, t_em = 0	t_ICE = 380, t_em = t_req - 380
380 < t_req <= 380+t_em_max	t_ICE = t_req, t_em = 0	t_ICE = 380, t_em = t_req - 380	t_ICE = 380, t_em = t_req - 380
t_req > 380+t_em_max	t_ICE = t_req, t_em = 0	t_em = t_em_max, t_ICE = t_req - t_em	t_em = t_em_max, t_ICE = t_req - t_em

Driving Modes – Hybrid Mode

Structured Pseudocode

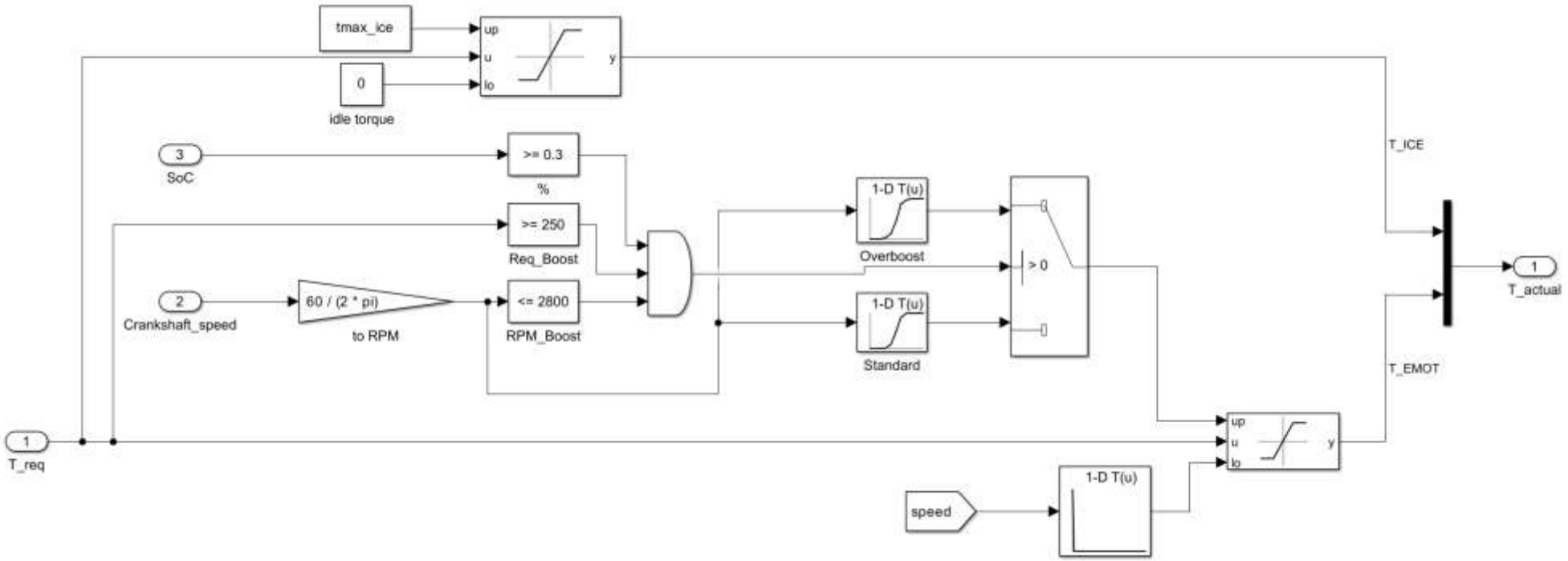
```
FUNCTION Engine_Sweetspot_Operation(t_req, t_em_max, SoC):
    IF SoC < 20% THEN    // Low-SoC Guard: Force Active Recharge (No EV Traction)
        IF t_req <= 0 THEN t_ICE = 0,  t_em = max(t_req, -t_em_max)
        ELSE IF t_req <= 380 THEN t_ICE = 380,  t_em = max(-t_em_max, t_req - 380)
        ELSE t_ICE = t_req,  t_em = 0    // Inhibit electric boost

    ELSE IF SoC > 95% THEN    // Overcharge Guard: Disable Battery Charging & Regen
        IF t_req <= 0 THEN t_ICE = 0,  t_em = 0    // Mechanical friction brakes only
        ELSE IF t_req <= 110 THEN t_ICE = 0,  t_em = t_req  // Pure EV Mode (Promote Discharge)
        ELSE IF t_req <= 380 THEN t_ICE = t_req,  t_em = 0    // ICE drives, zero recharge

    ELSE    // Normal Supervisory Hybrid Strategy (20% <= SoC <= 95%)
        IF t_req <= 0 THEN t_ICE = 0,  t_em = max(t_req, -t_em_max)    // Regen Braking
        ELSE IF t_req <= 110 THEN t_ICE = 0,  t_em = t_req    // Pure EV Mode
        ELSE IF t_req <= 380 THEN t_ICE = 380,  t_em = max(-t_em_max, t_req - 380)//Sweet-Spot
```

Driving Modes – Race Mode

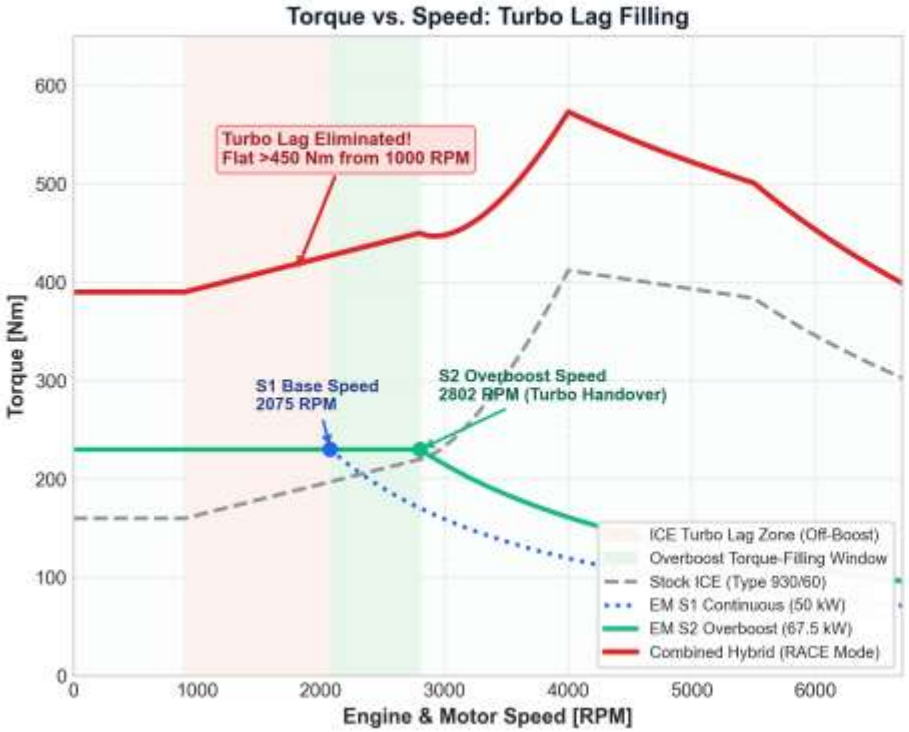
Simulink Scheme:

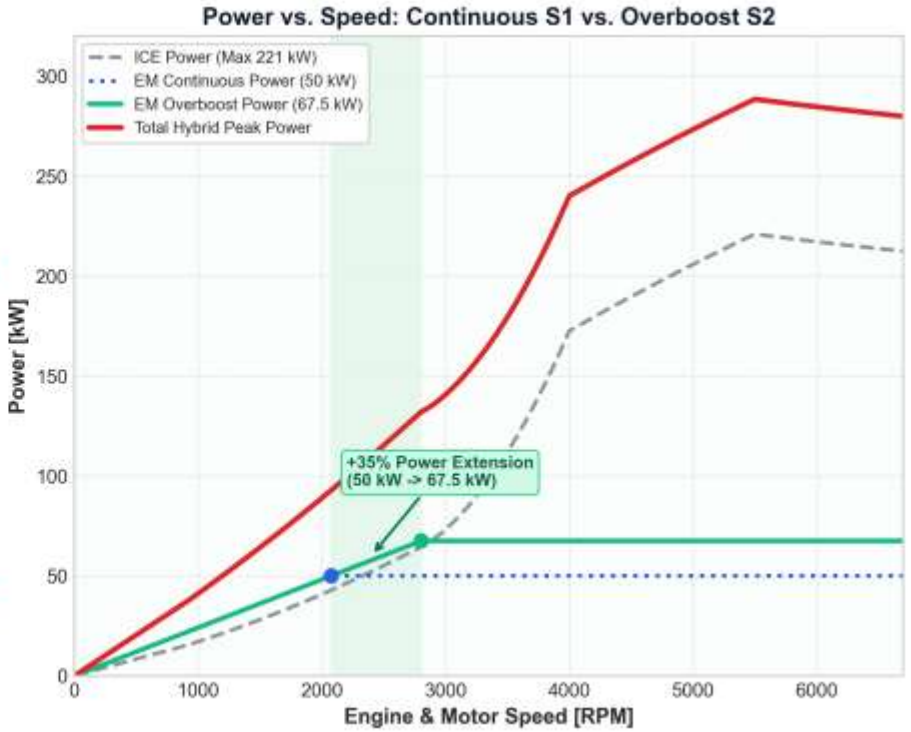


Race Mode:

This Mode operates without any sweetspot control for maximizing the performance output. The torque request from the both ICE and Electric Motor is saturated (if the battery allows it) using respective torque curves and allowing the **overboost control** of the EM.

ECU – EM Overboost





4. Battery Feasibility Verified

Can our 2.9 kWh pack deliver it?

- Molicel P42A 21700: Rated for 45 A/cell continuous (10.7C rate).
- Simulated Peak on Track: 74.5 A pack (37.25 A/cell) — well inside the 45 A cell limit.
- Voltage Margin: Stays at 366 V (min cutoff is 280 V)

1. Speed / Power Extension

Not a torque overload

Peak torque remains capped at 230 Nm (governed by stator current Iq ≈ 139 Arms). The +35% power boost (50 -> 67.5 kW) overcomes rising back-EMF, extending maximum torque to higher RPM.

2. Targeted Turbo Handover

Why exactly 2800 RPM?

The KKK K27 single turbo builds boost (0 -> 0.8 bar) starting at 2800 RPM. Overboost extends the 230 Nm plateau from 2075 to 2802 RPM, perfectly handing over full torque to the pressurized ICE.

3. ECU Trigger Logic

Race Mode Only

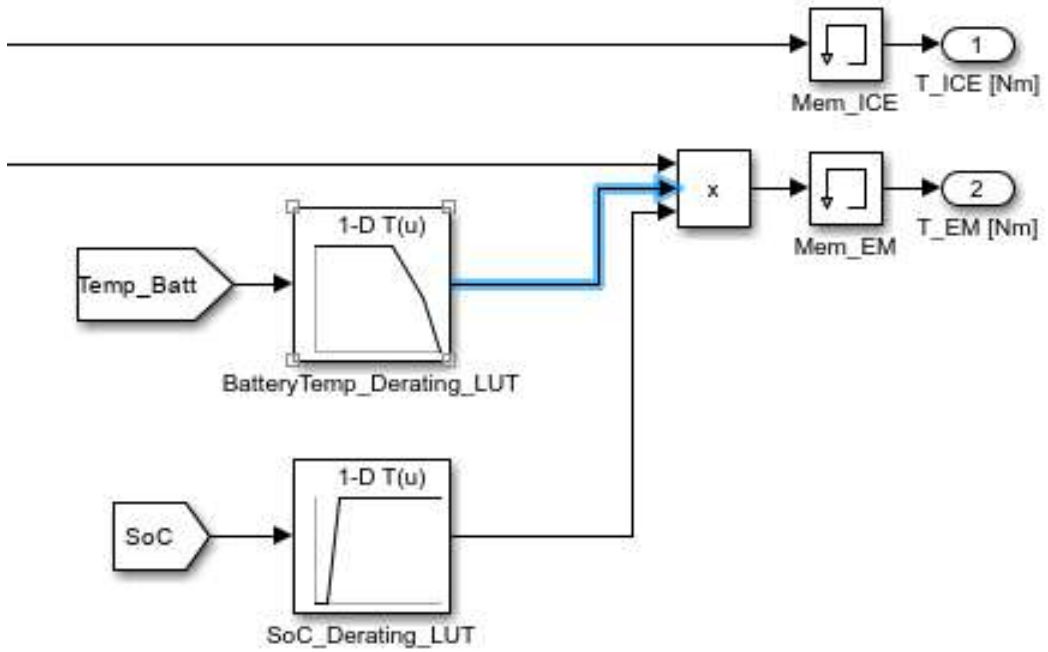
Automatically activated when:

- T_req ≥ 250 Nm (High driver demand)
- RPM ≤ 2800 (Turbo lag window)
- SoC ≥ 30% (Battery safe zone)

Disengages once turbo spool-up completes.

ECU – Torque Deratings

Simulink Scheme:



Torque Deratings:

For safe operations, the torque requests for the EM is derated using LUTs. The first LUT derates the torque for high battery temperatures, above 55 Celsius, and the second LUT derates the torque for low SoC, below 20%.



Results – Imola Lap Time Simulation

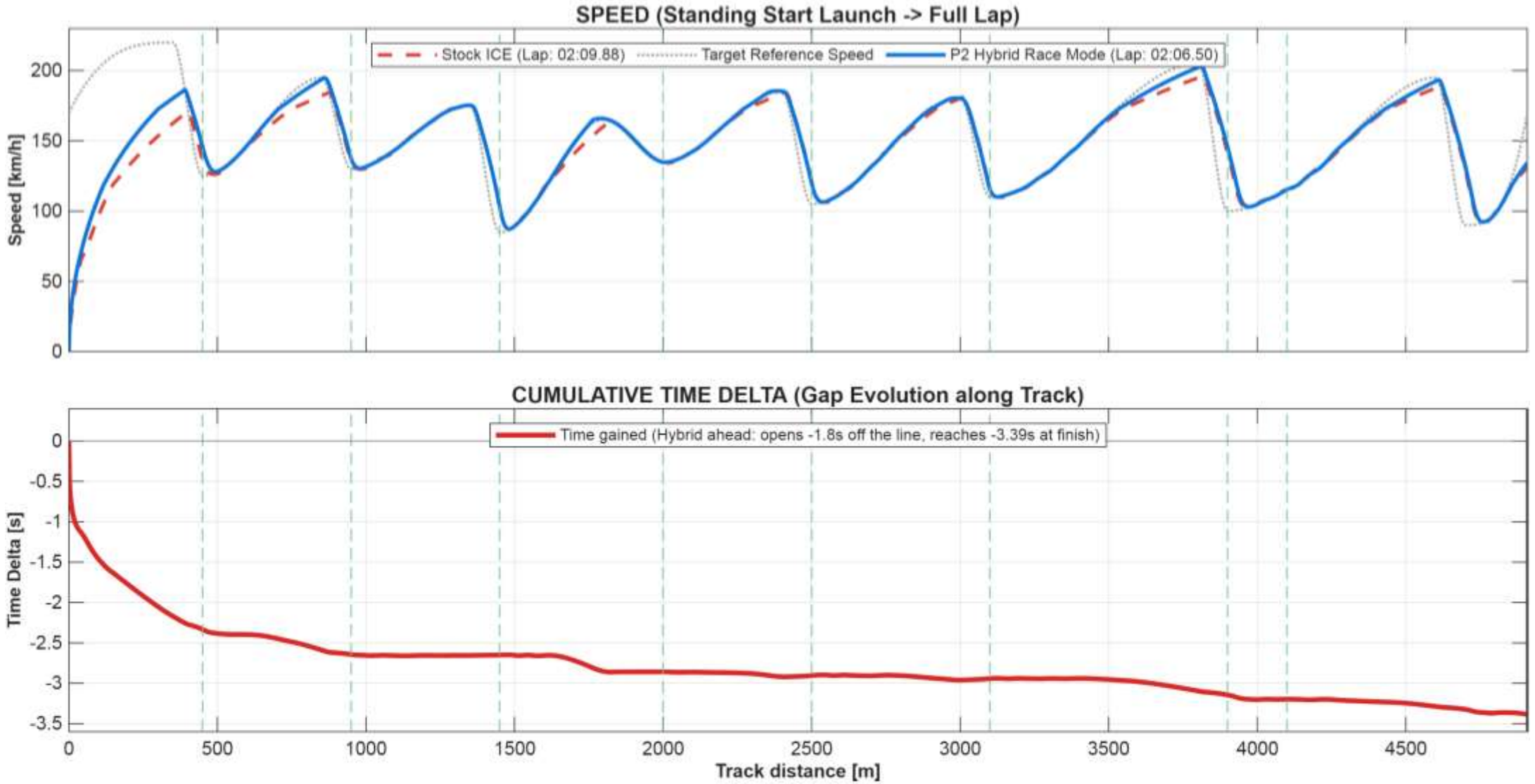
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EMILIA-ROMAGNA

Results – Imola Lap

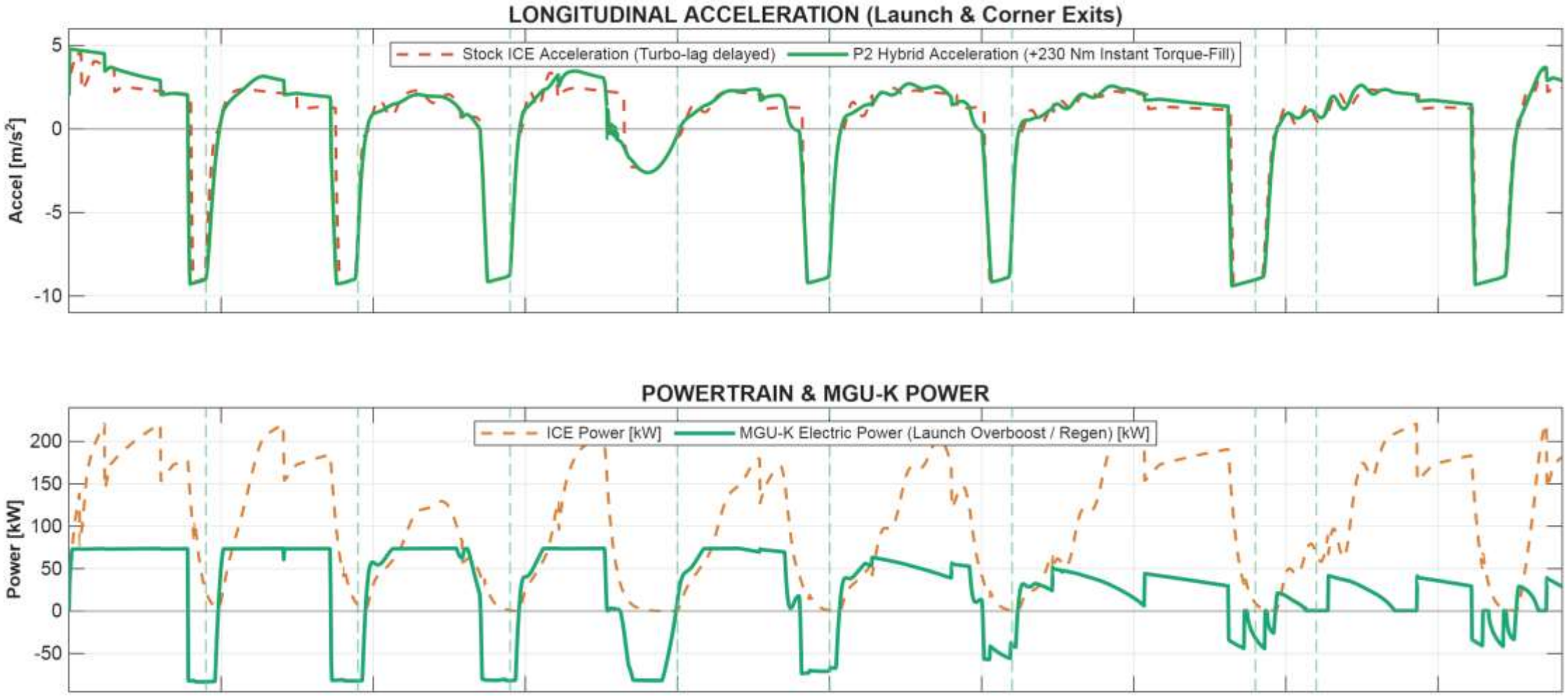
- **Stock ICE vs P2 HYBRID – Standing Start**



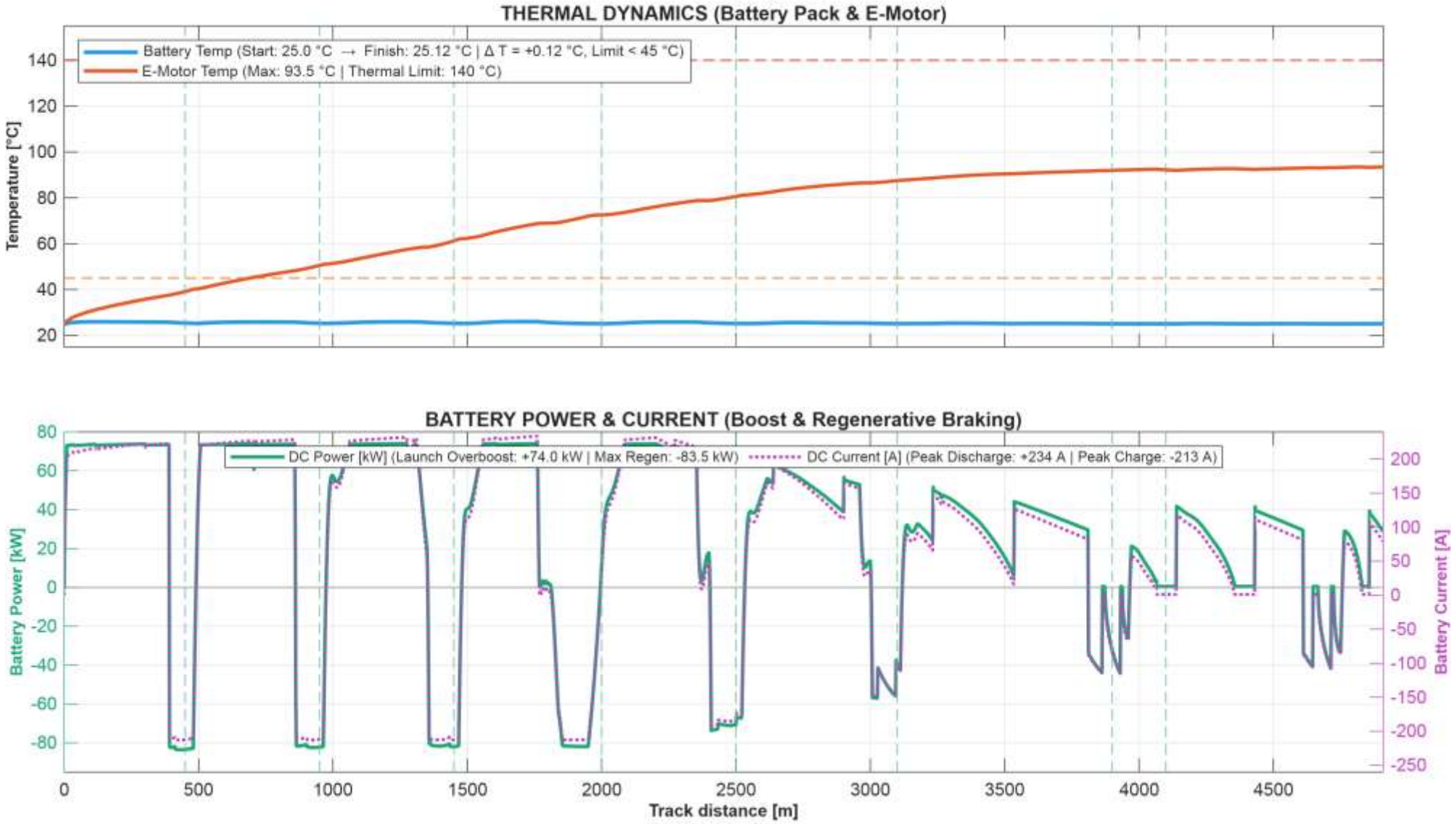
Results – Imola Lap



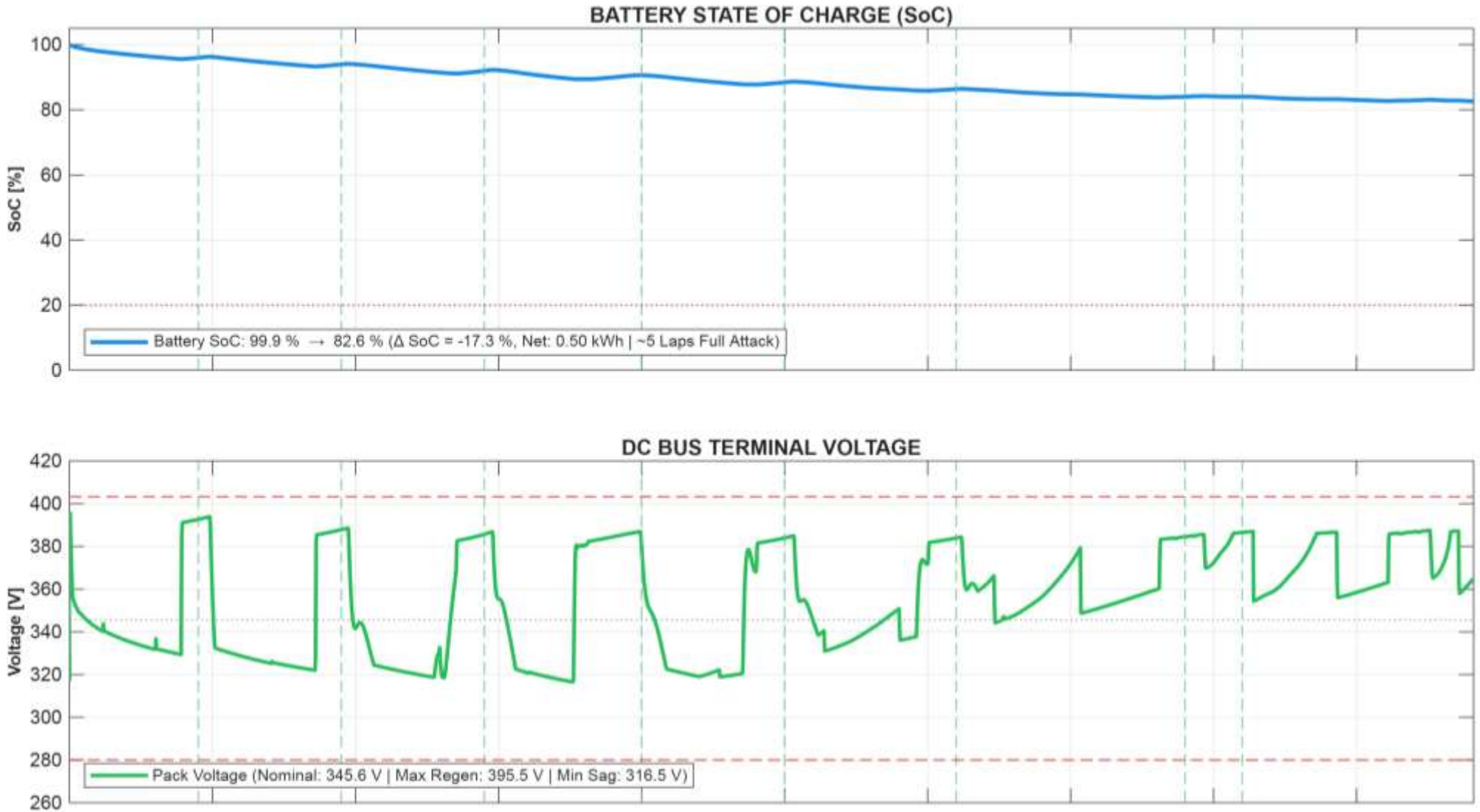
Results – Imola Lap



Results – Imola Lap



Results – Imola Lap

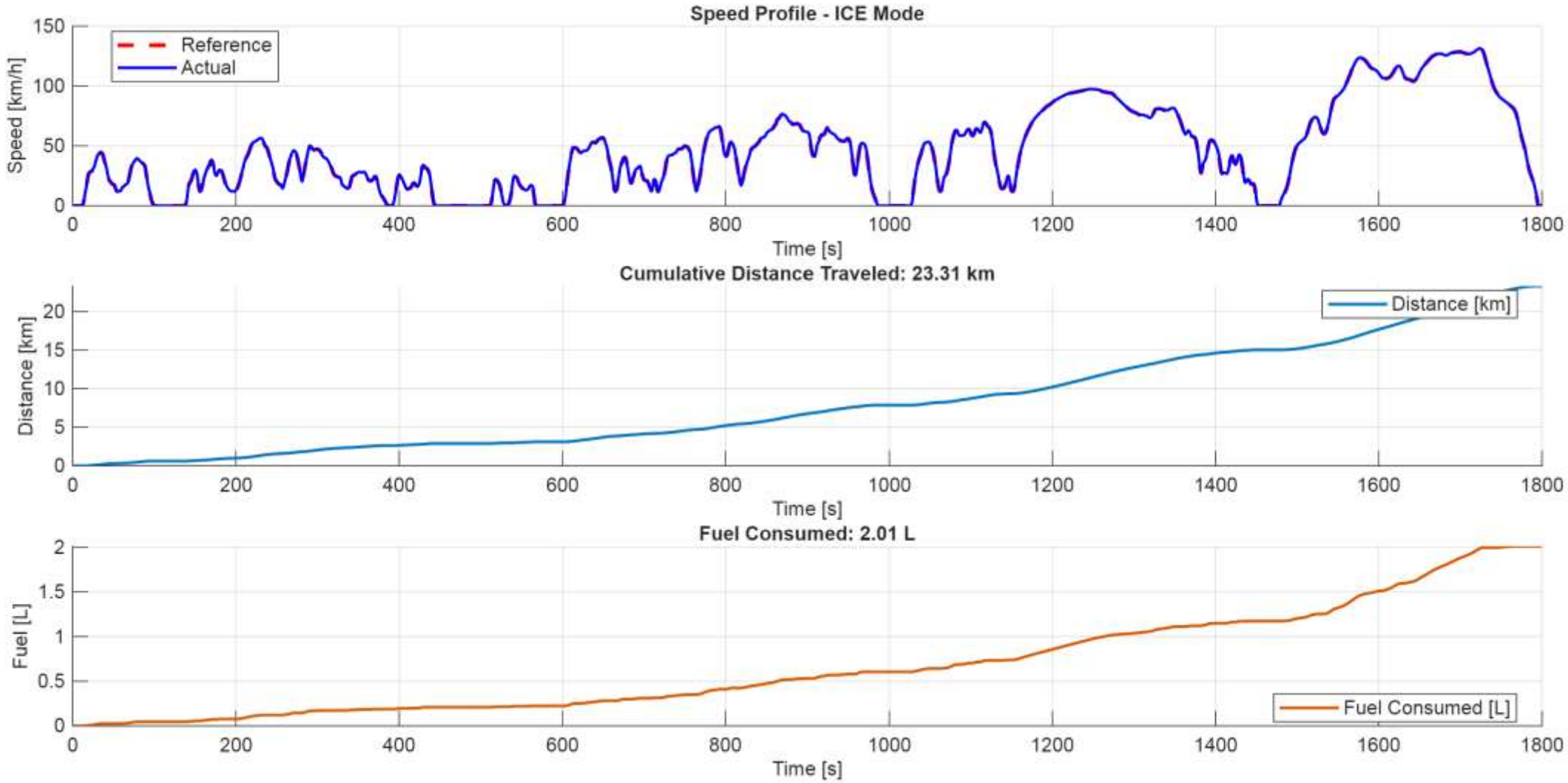


Results – WLTP Class 3b

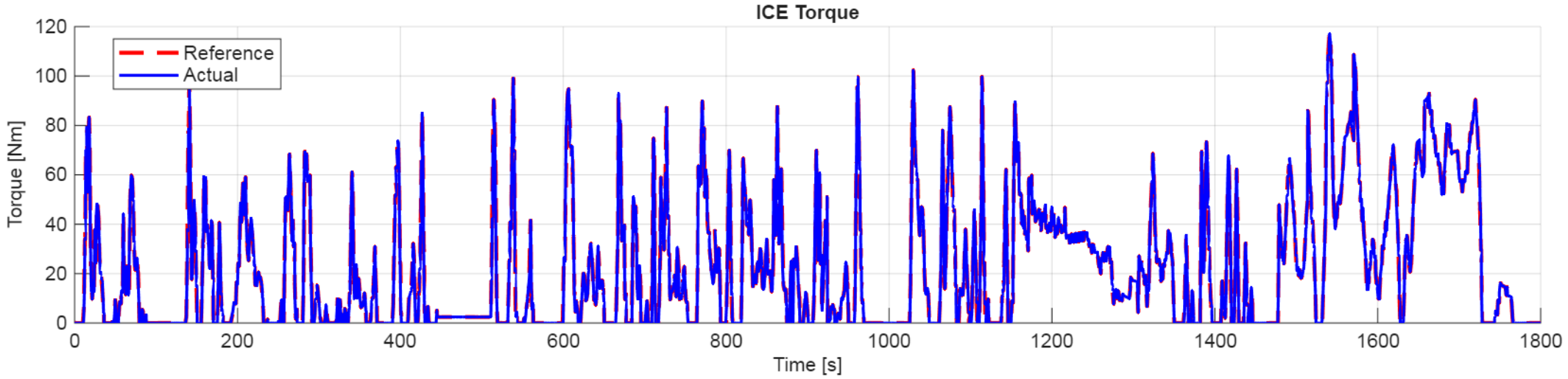


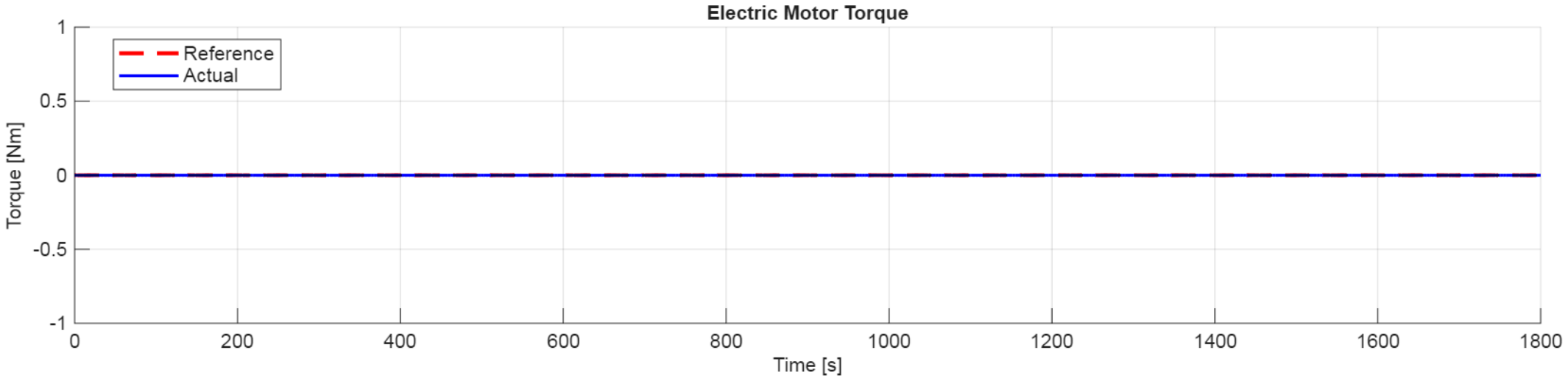
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Results – WLTP Class 3b - ICE Mode

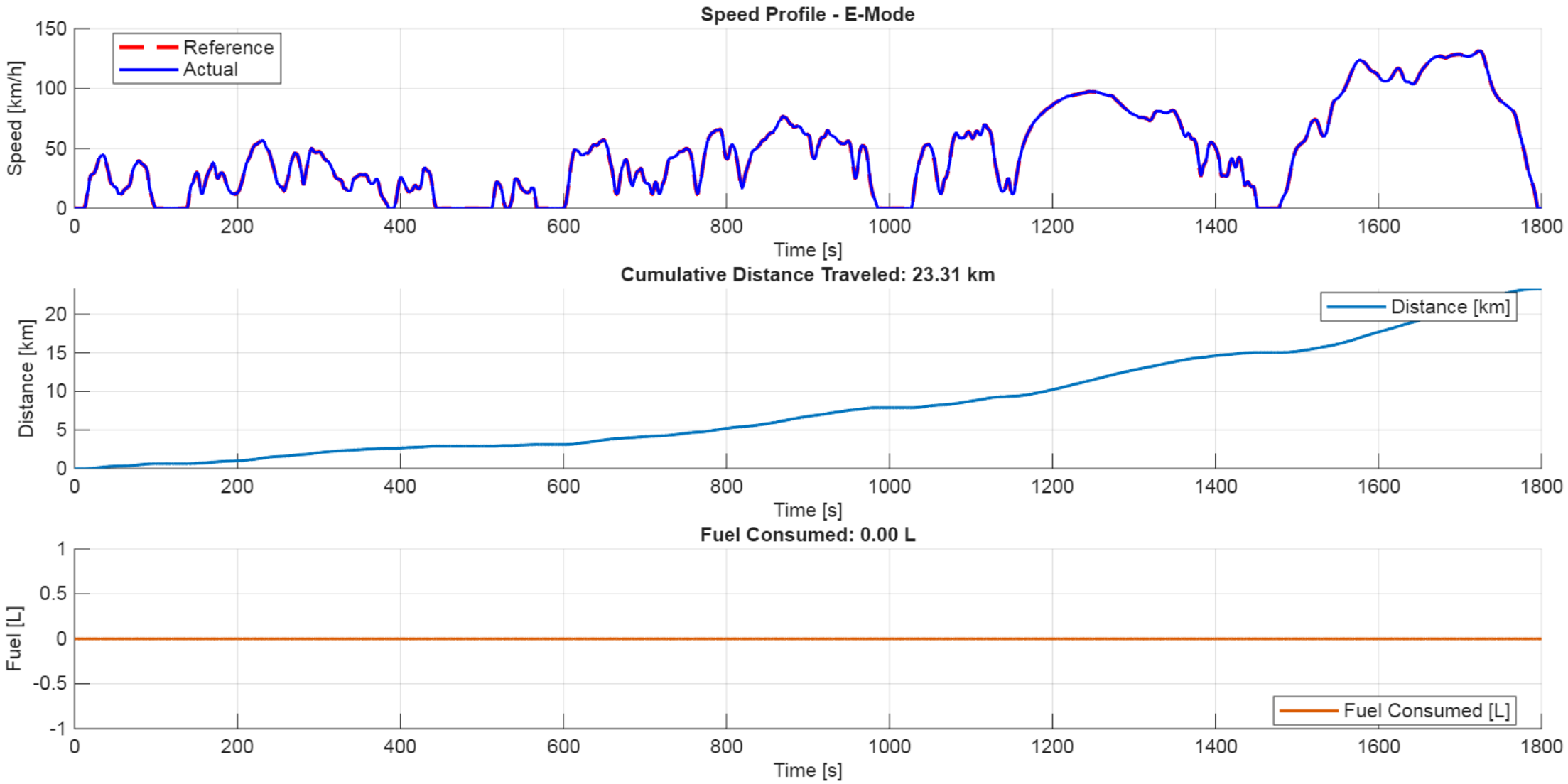


Results – ICE Mode – Torque Distribution

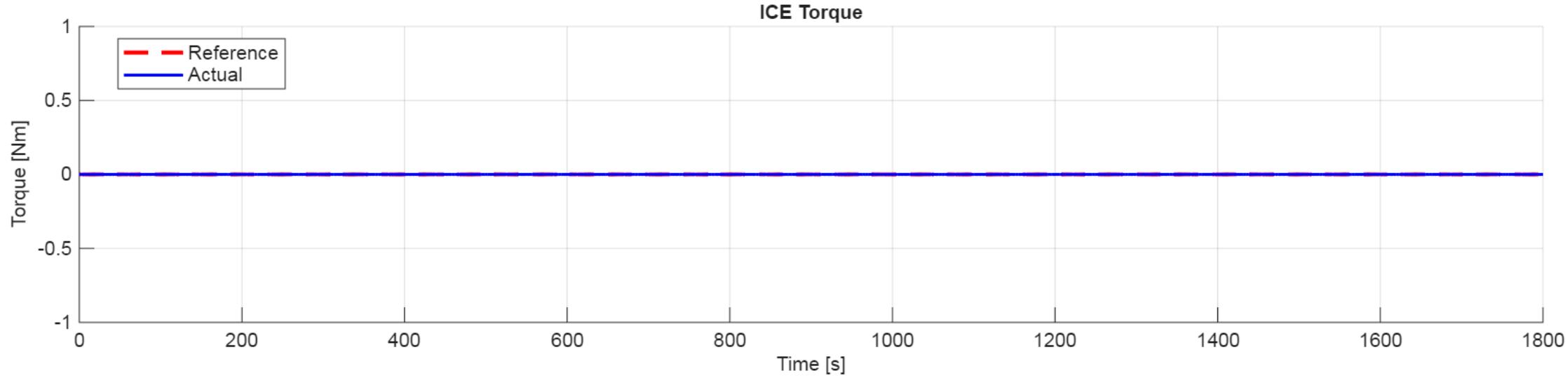


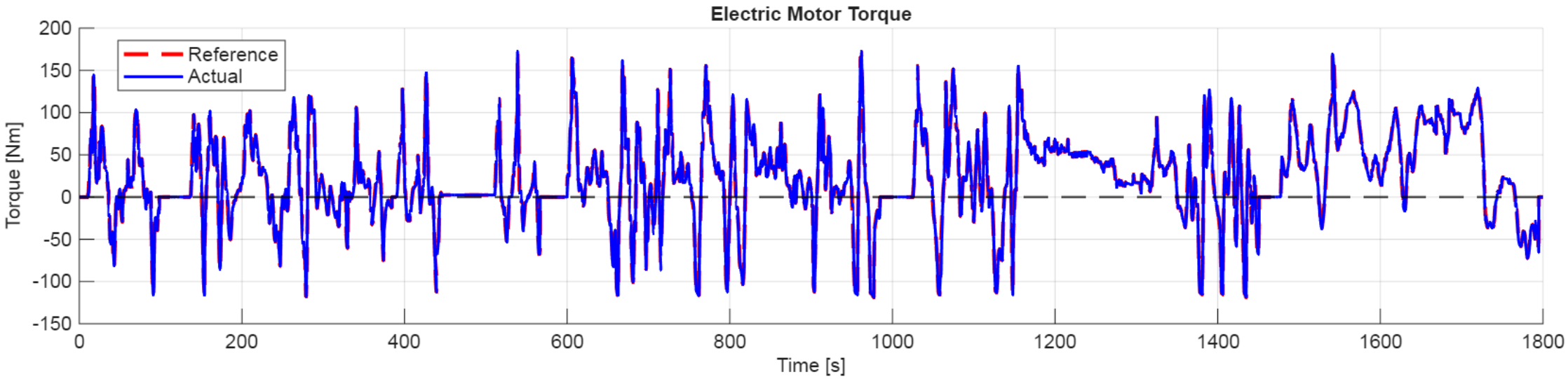


Results – WLTP Class 3b - Emode

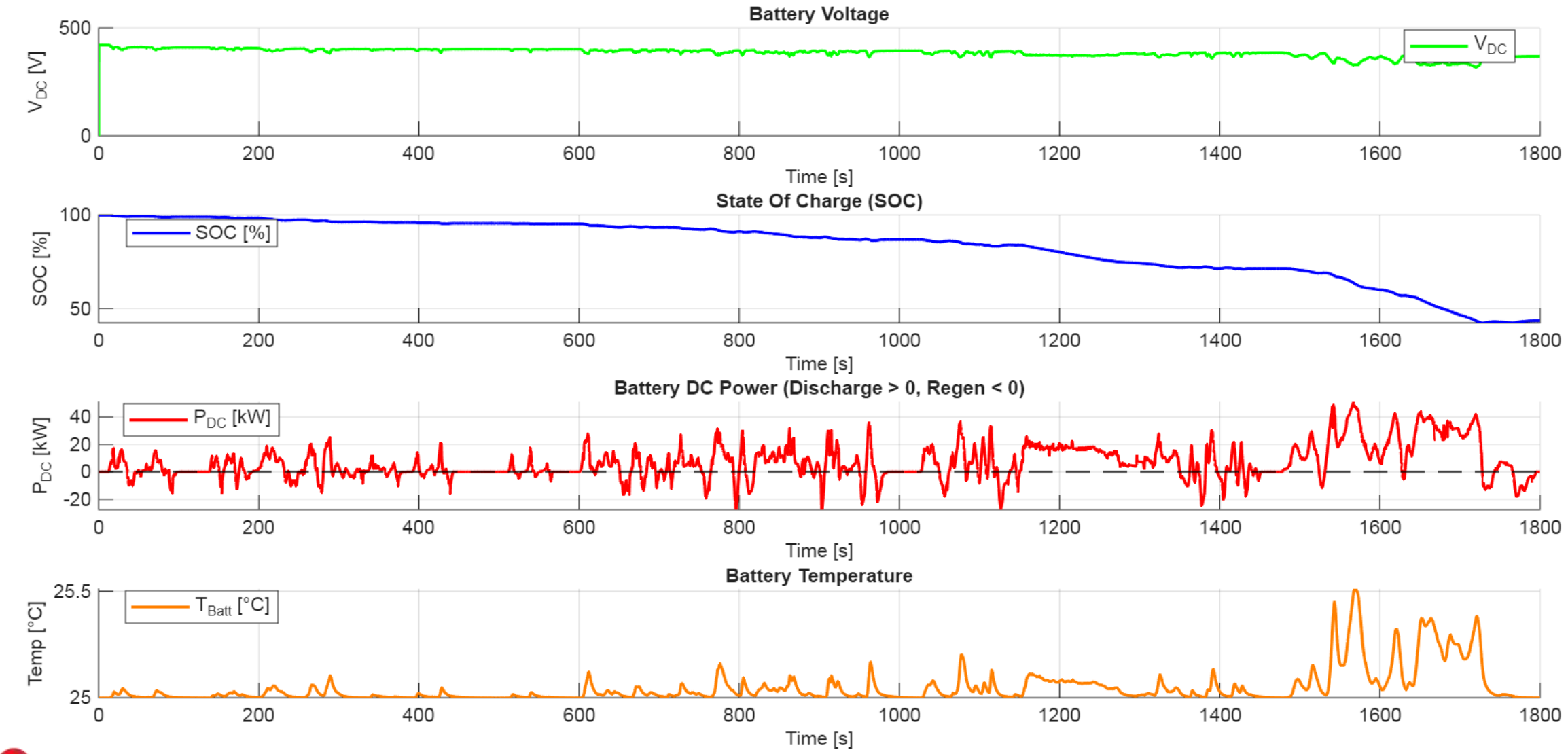


Results – EMode – Torque Distribution

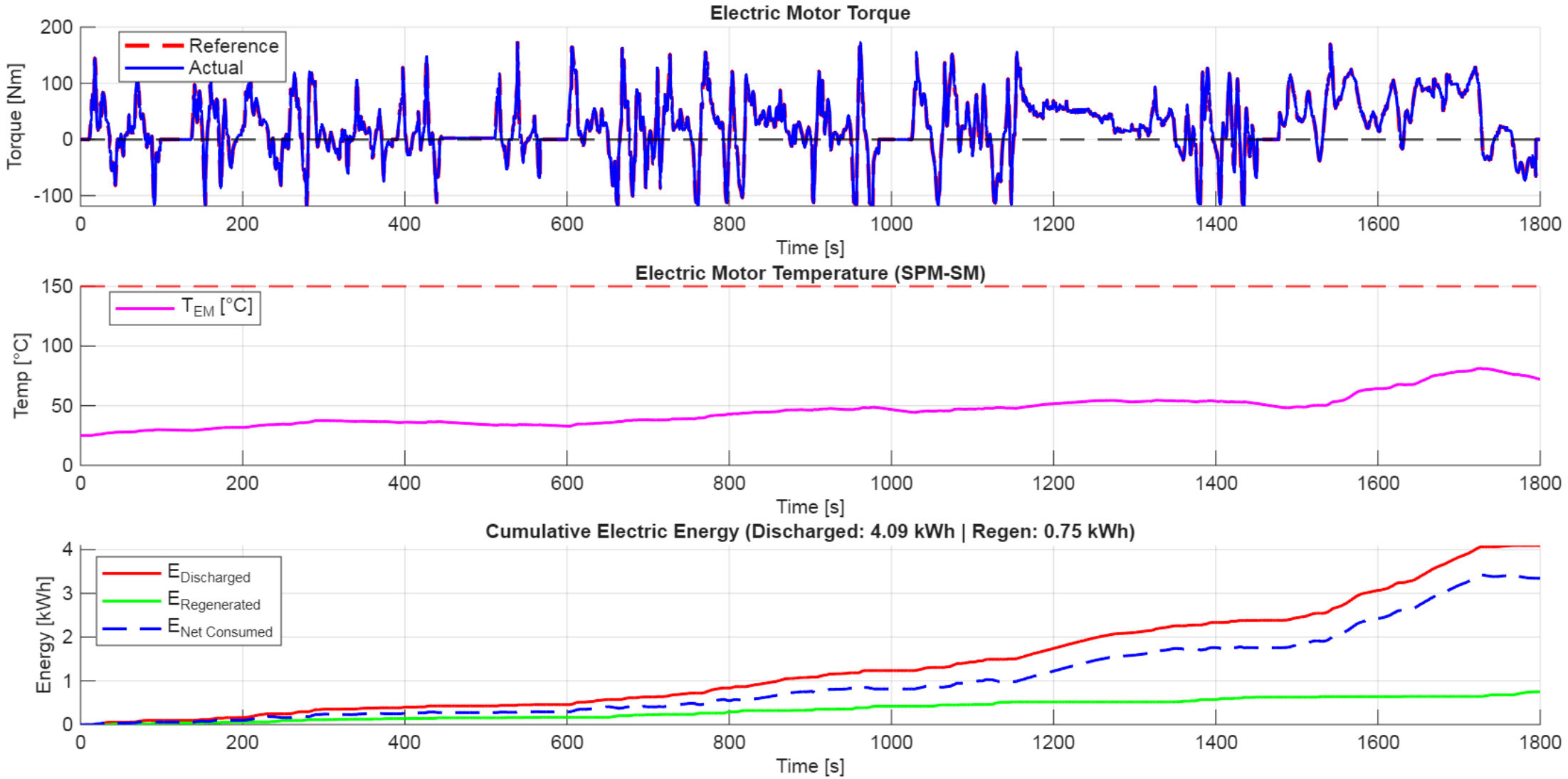




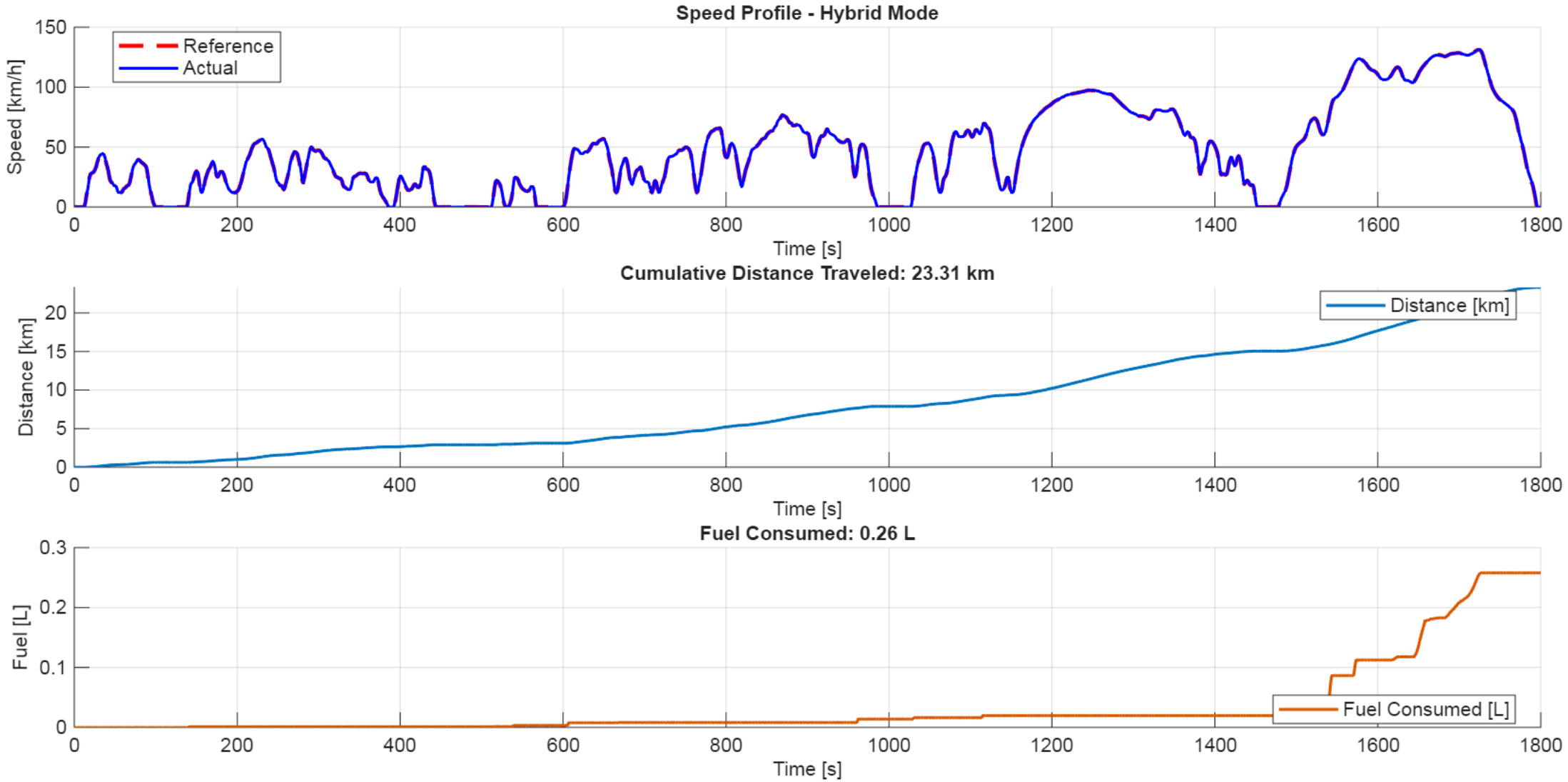
Results – EMode – Battery Characteristics



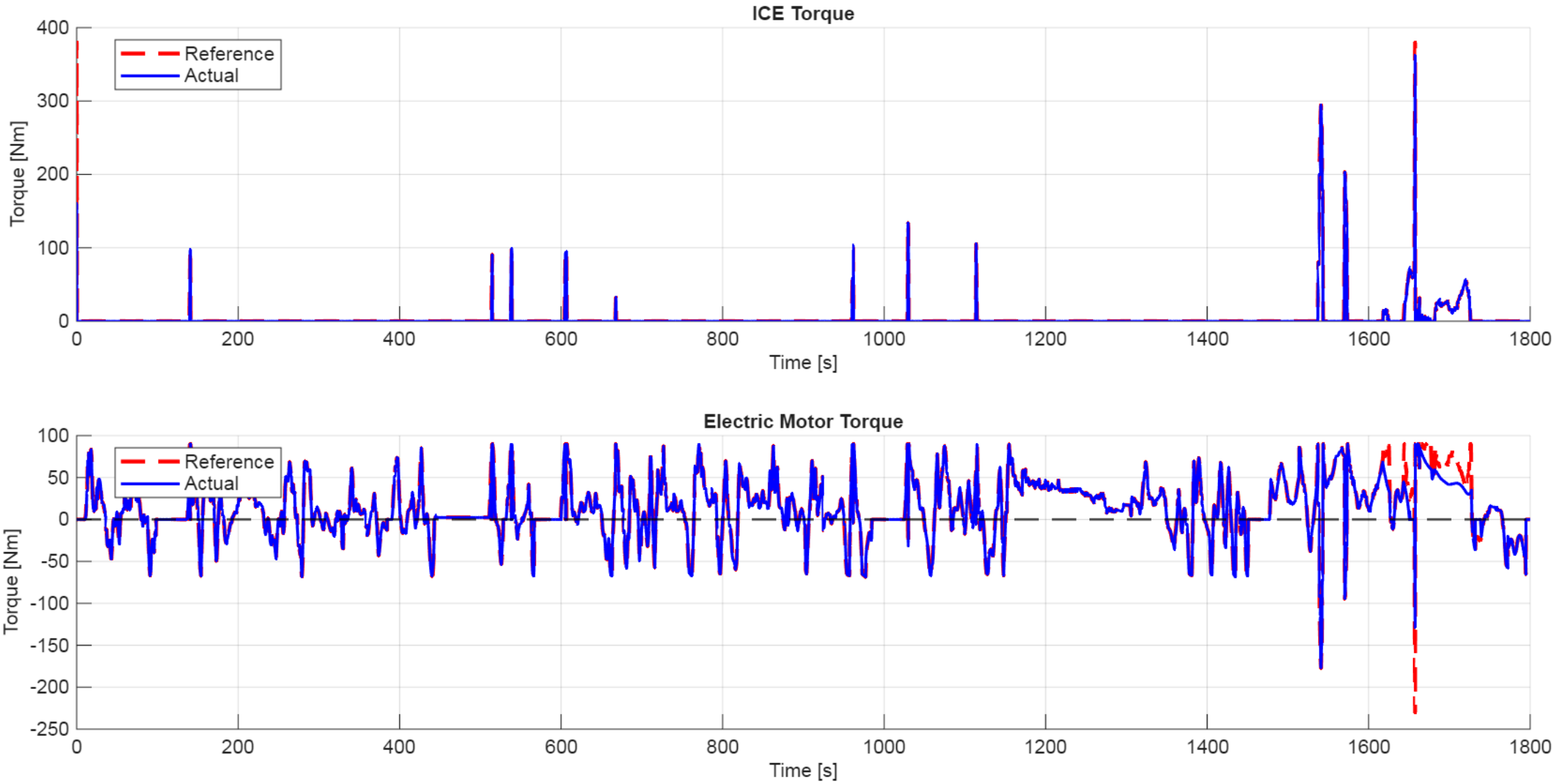
Results – EMode – Energy Delivery



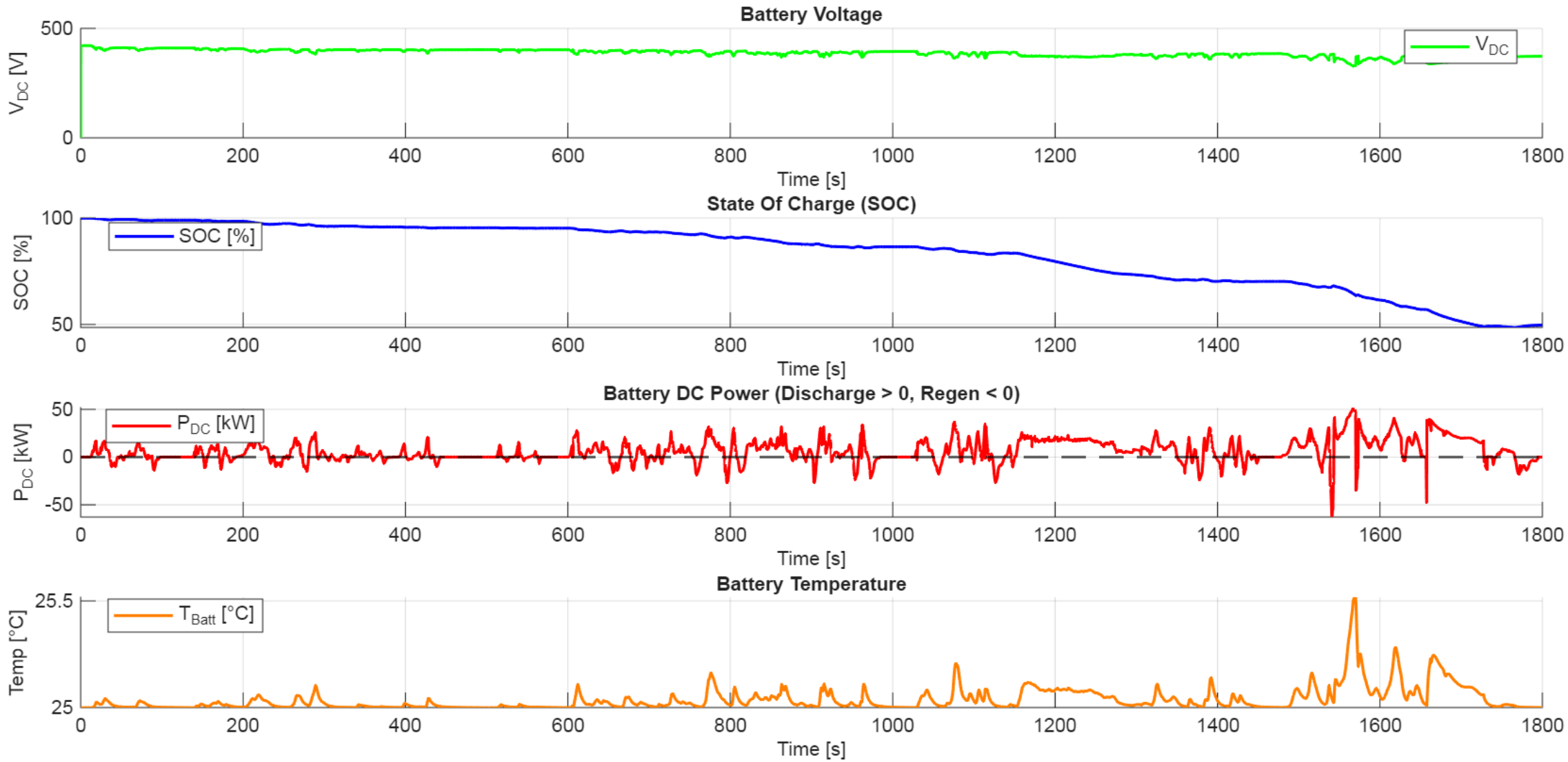
Results – WLTP Class 3b - Hybrid Mode



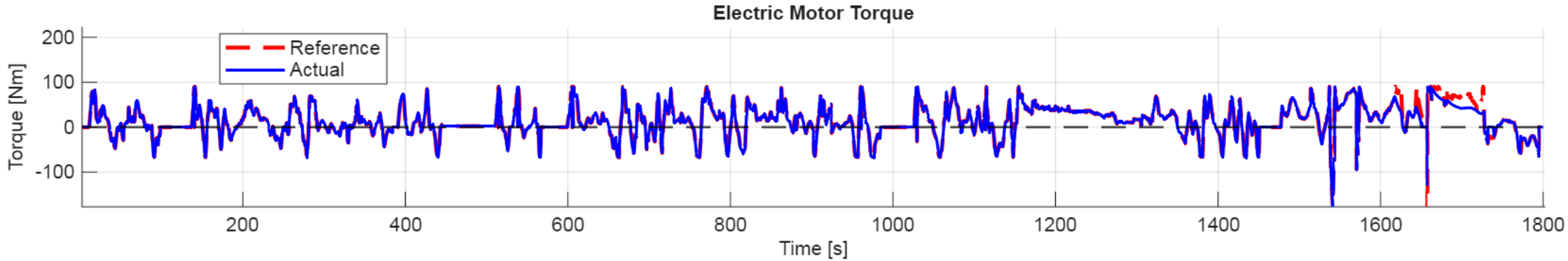
Results – Hybrid Mode – Torque Distribution

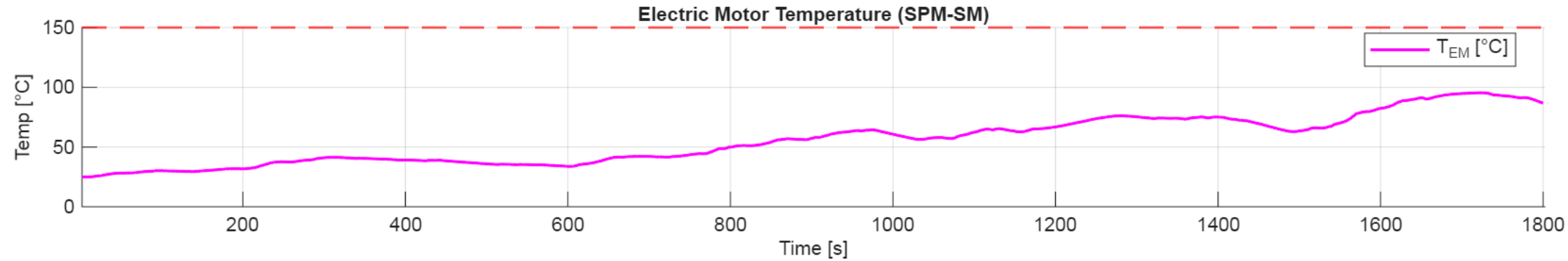


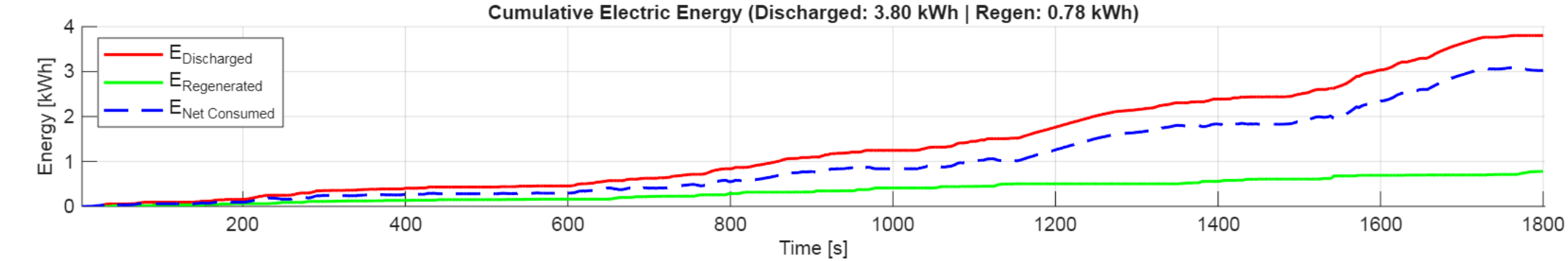
Results – Hybrid Mode – Battery Characteristics



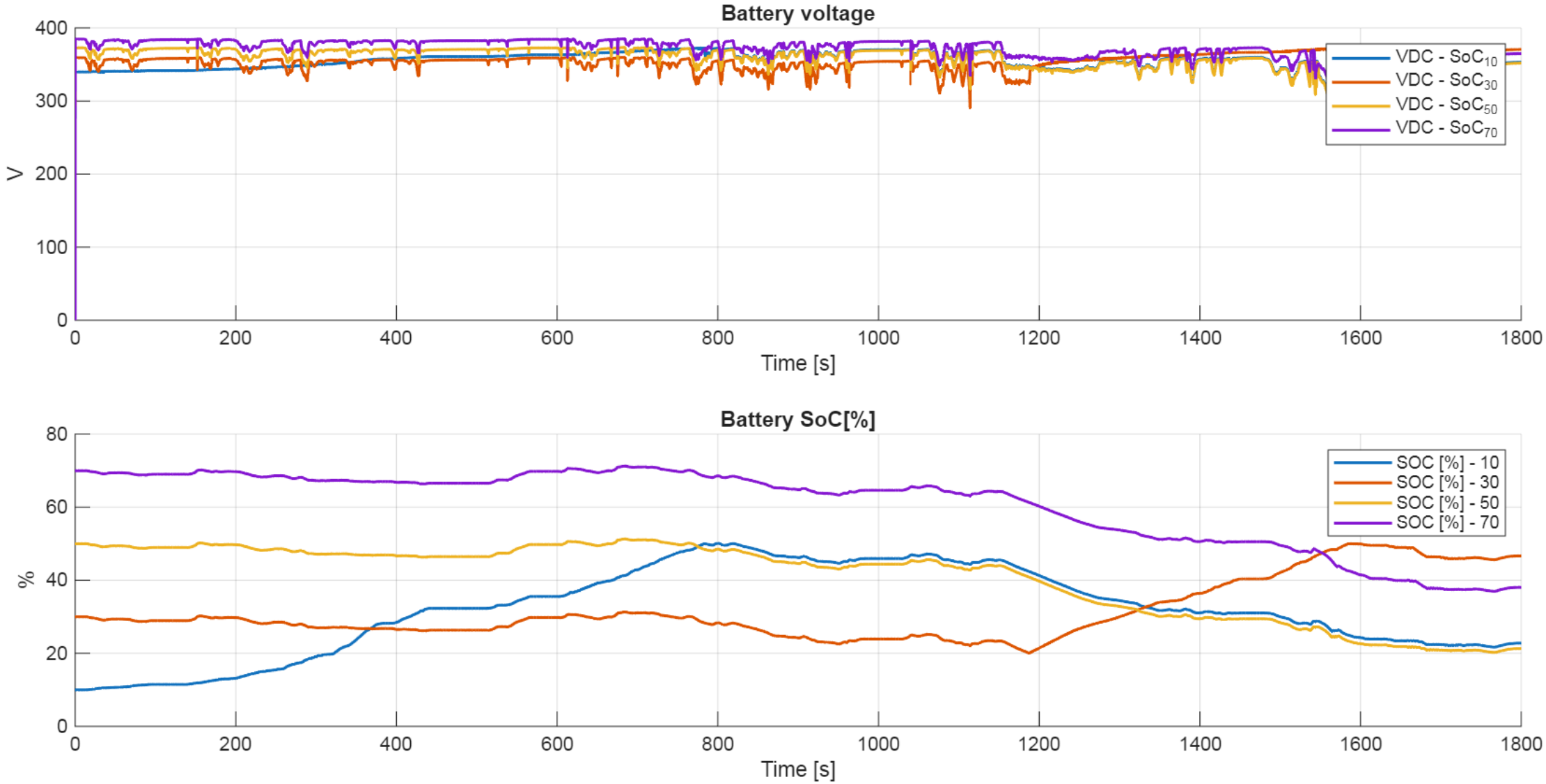
Results – Hybrid Mode – Energy Delivery



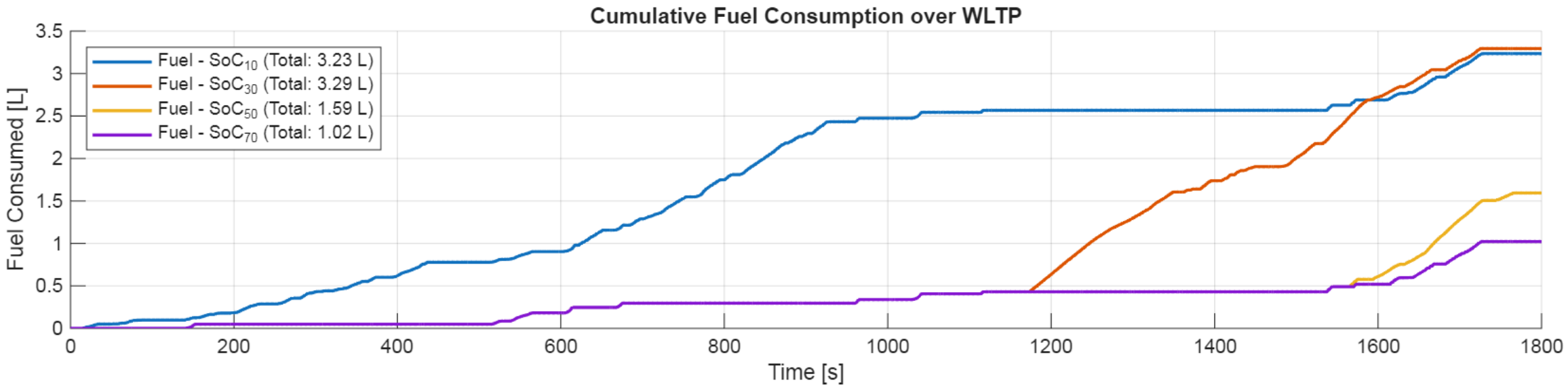




Results – Hybrid Mode – at different SoCs



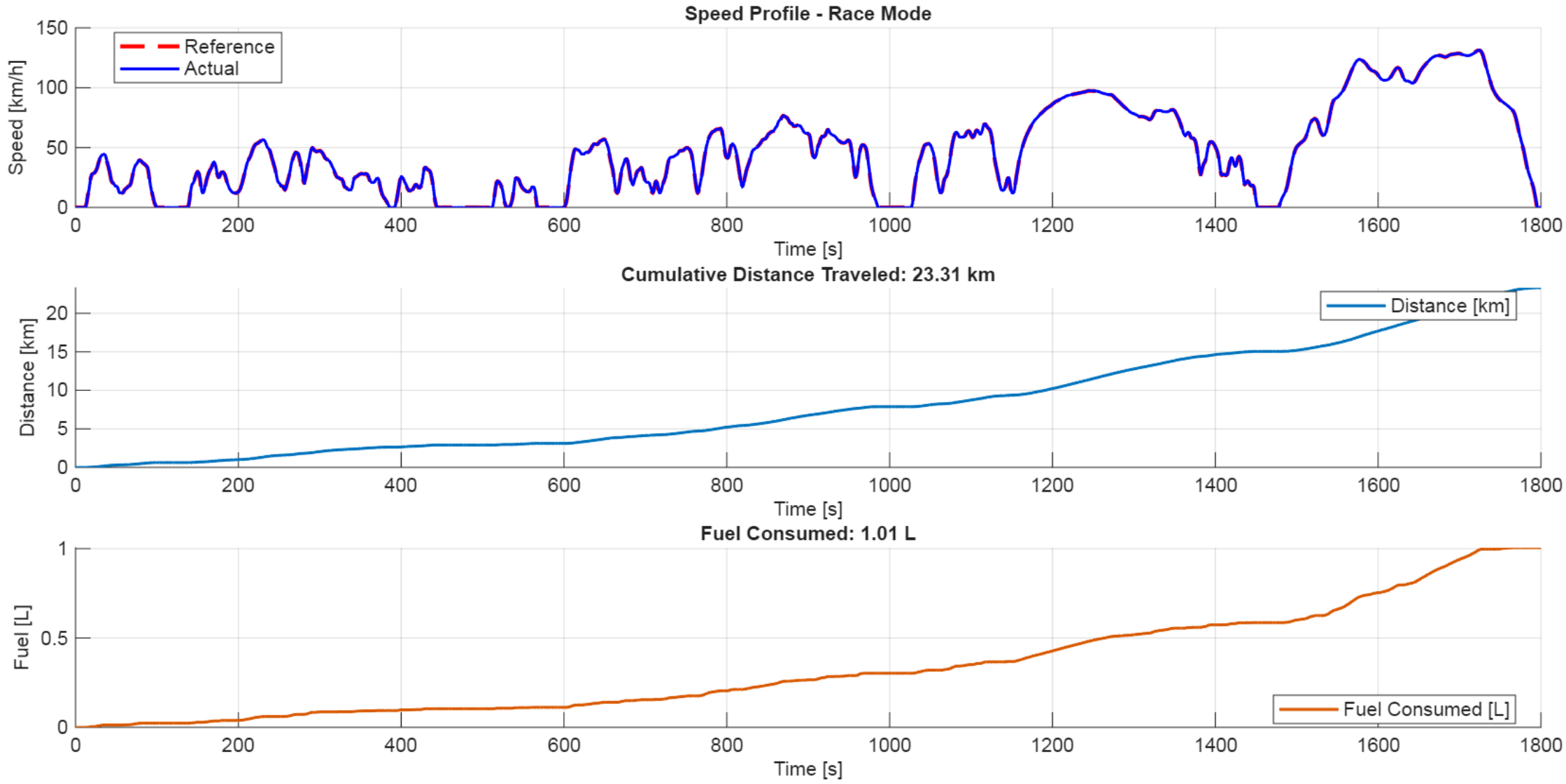
Results – Hybrid Mode – at different SoCs



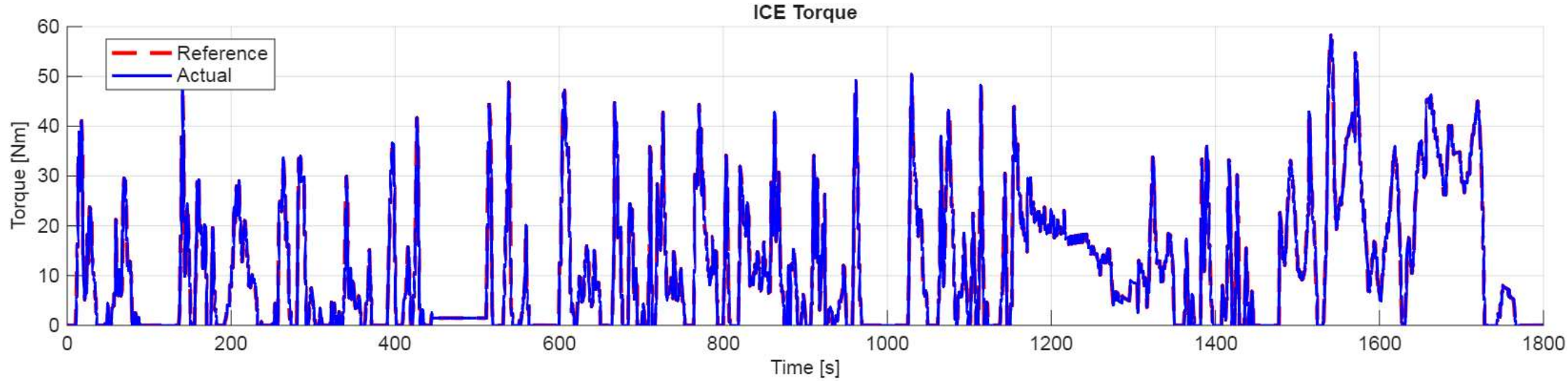
Why with a higher initial battery charge at **SoC 30%**, it registers a slightly higher cumulative fuel consumption than at **SoC 10% (3.29 L vs 3.23 L)?**

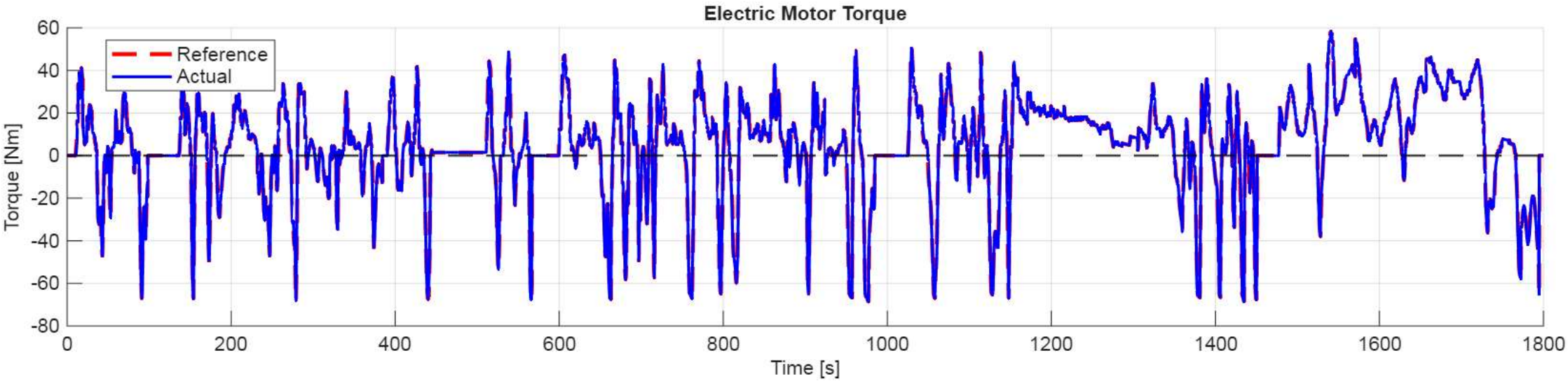
- At SoC 10%, forced recharge occurs early ($t = 0 - 950$ s) during low-speed urban phases where road load is minimal, enabling efficient battery replenishment at low-to-medium engine speeds before stabilizing at the 50%.
- At SoC 30%, pure EV depletion delays the 20% trigger. As a result, forced recharge coincides with the **WLTP Extra-High speed phase** *(100–130 km/h).* Here, the engine must simultaneously overcome high aerodynamic resistance and supply maximum generator charging torque at higher RPMs, shifting the ICE into operating points with higher mechanical losses and slightly elevated specific fuel consumption.

Results – WLTP Class 3b - Race Mode

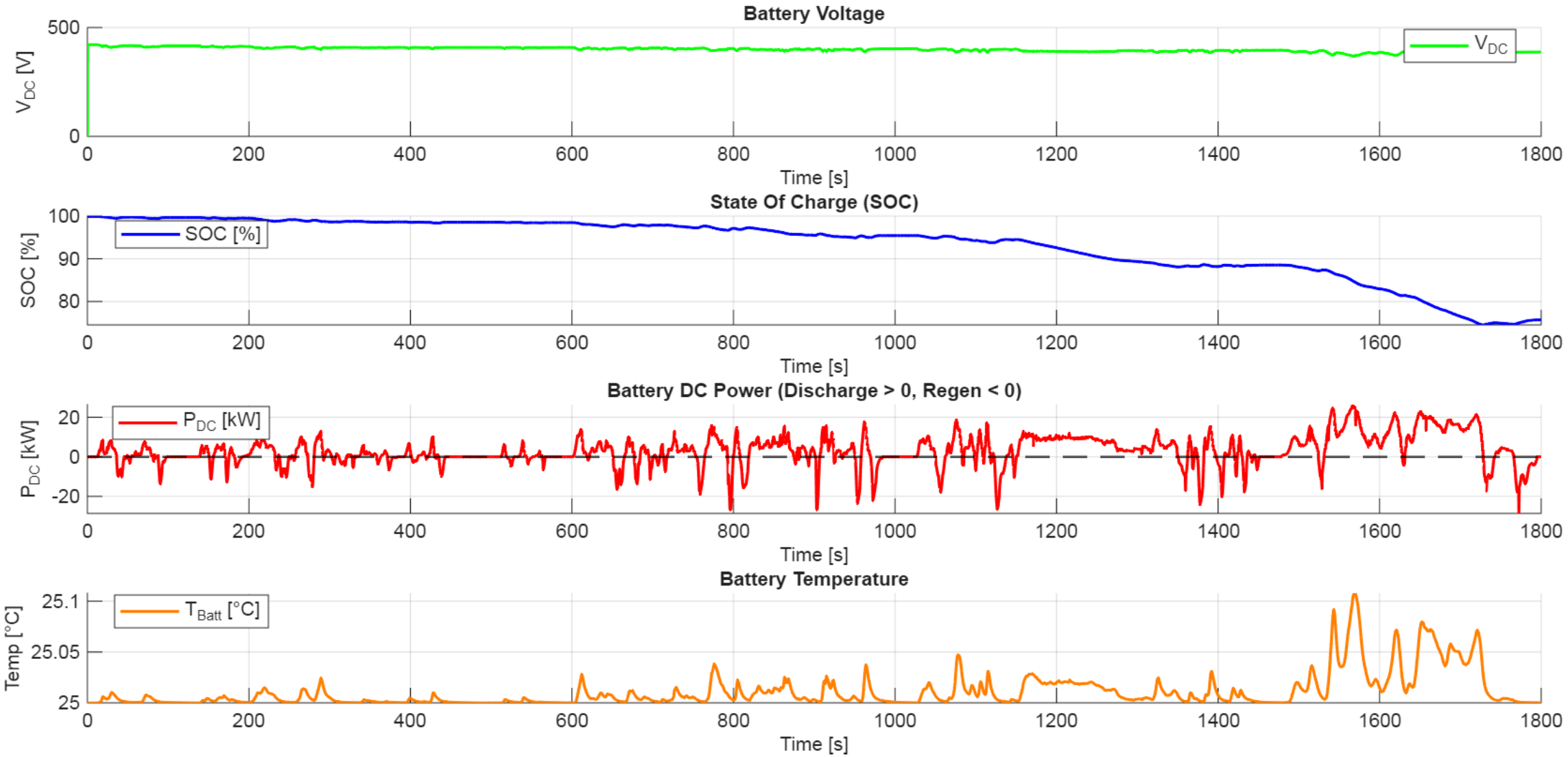


Results – Race Mode - Torque Distribution

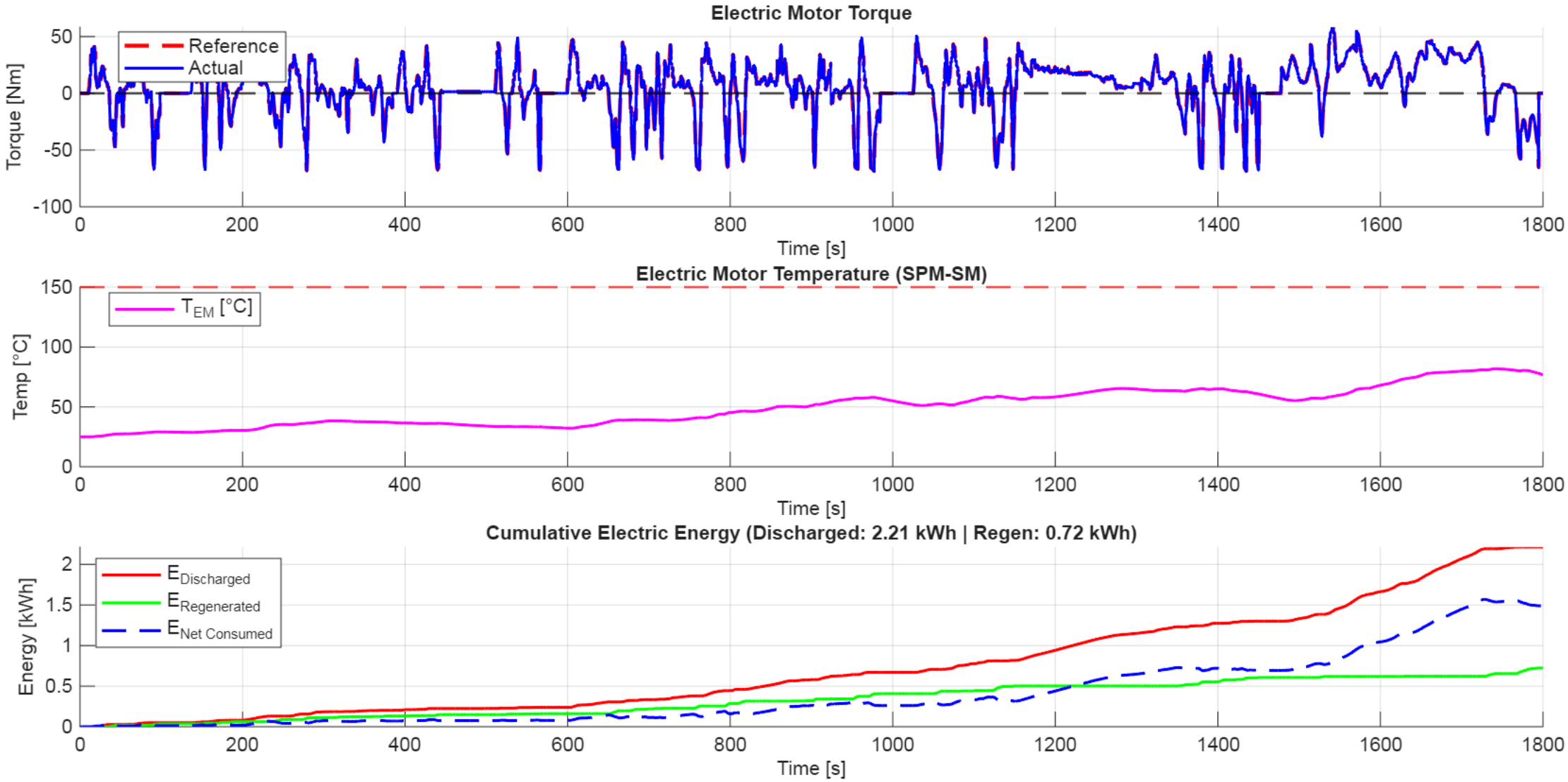




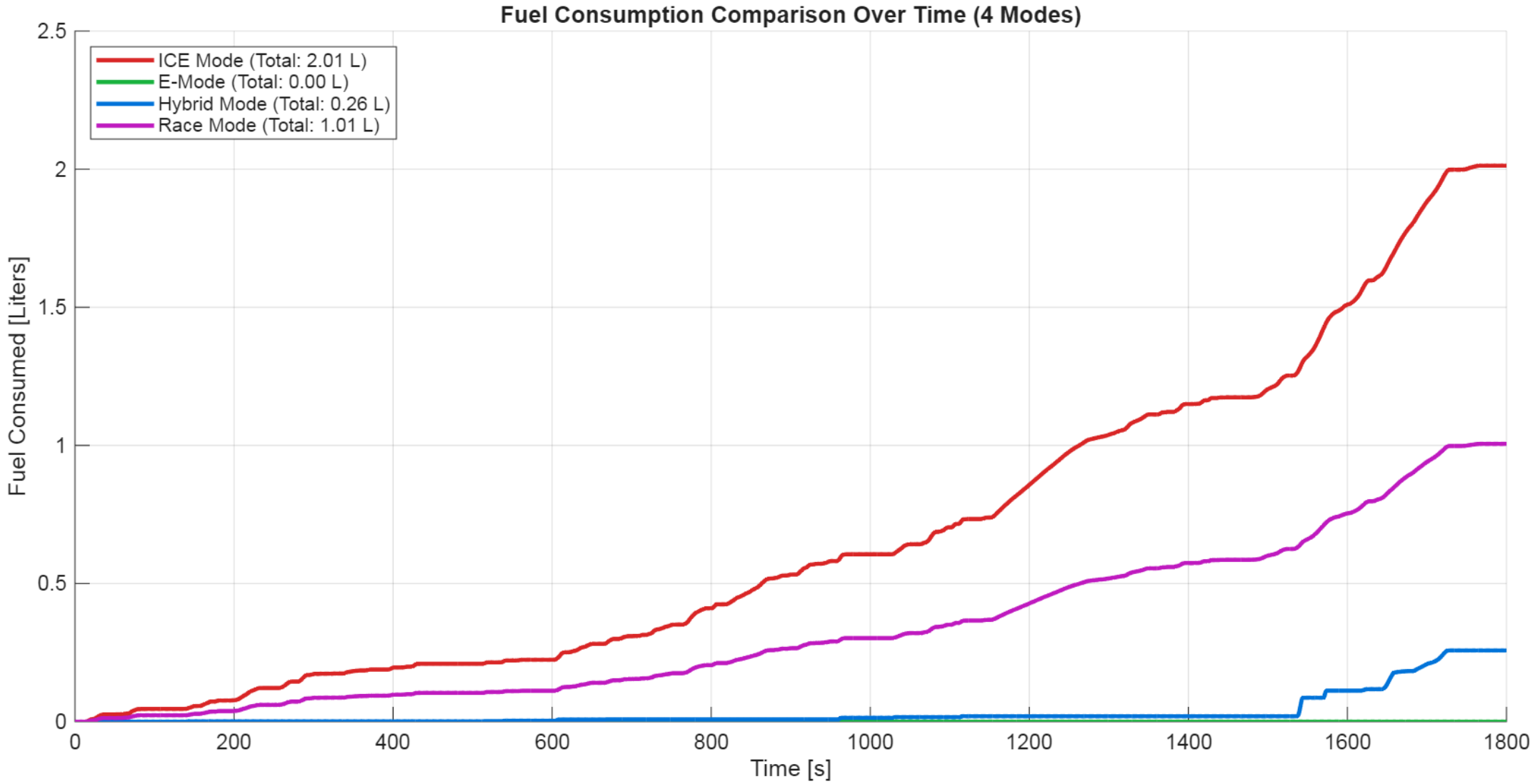
Results – Race Mode - Battery Characteristics



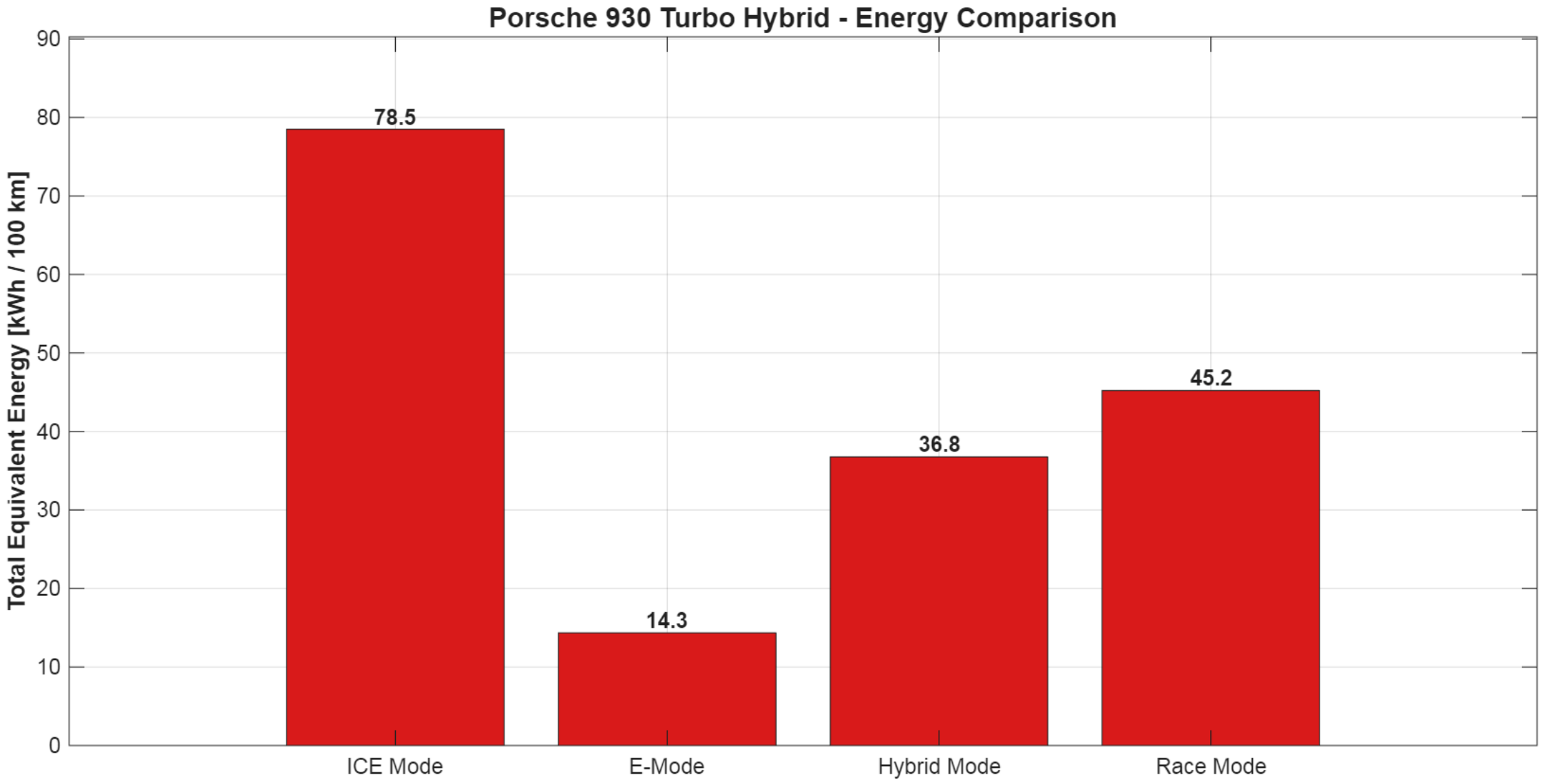
Results – Race Mode – Energy Delivery



Results – Fuel Consumption



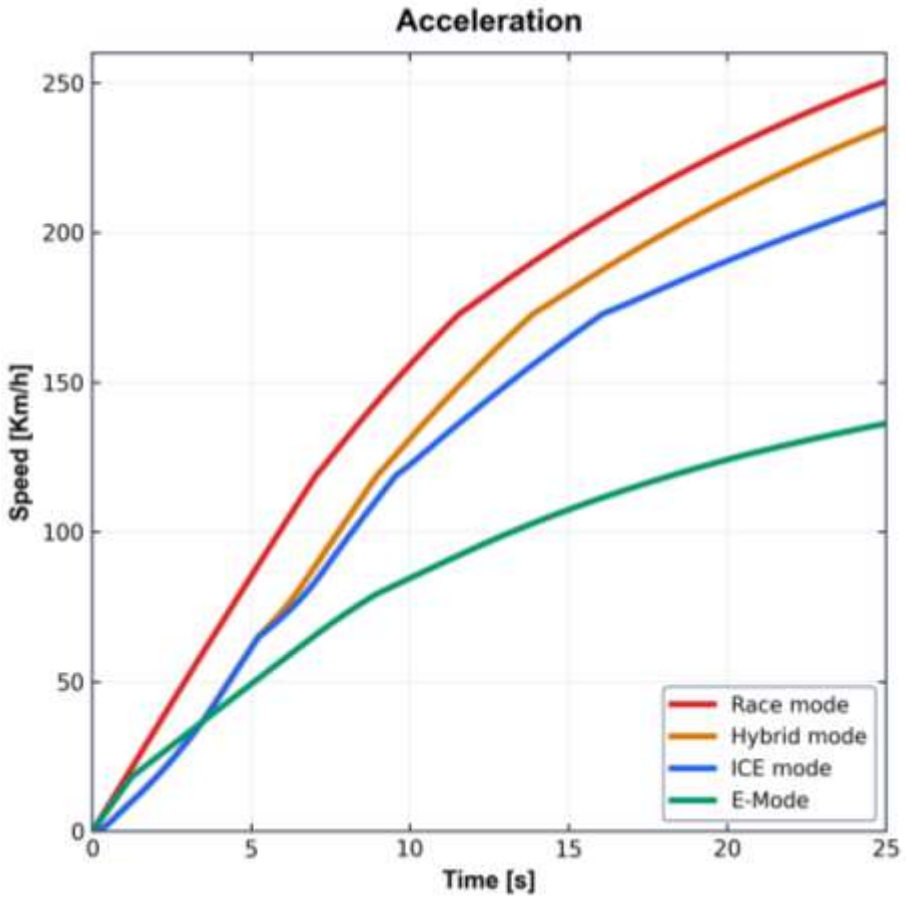
Results – Energy Consumption



Overall Results – Acceleration

ICE mode	
Speed [Km/h]	Time [s]
100	8.2
200	22.3

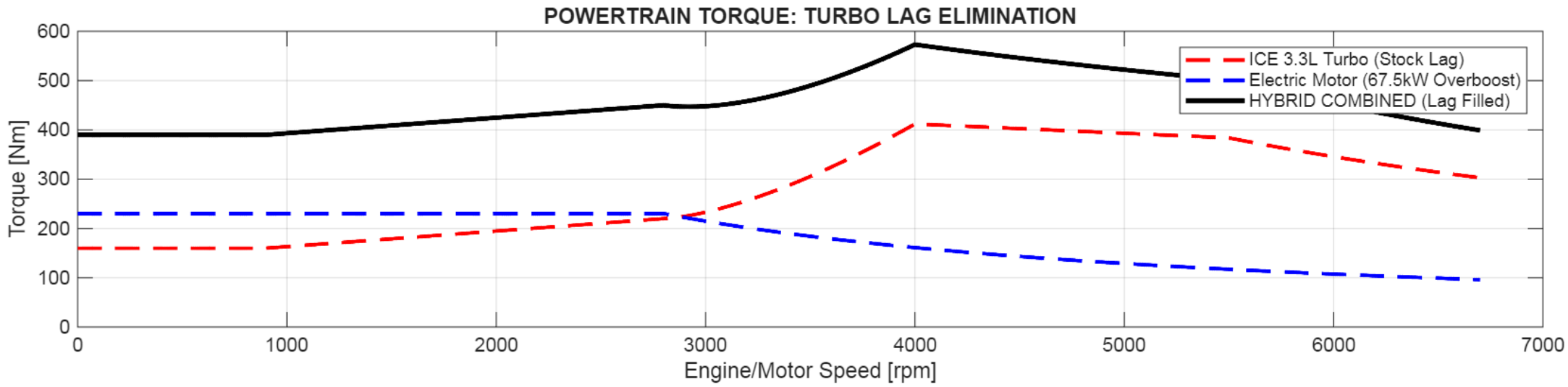
Race mode	
Speed [Km/h]	Time [s]
100	5.9
200	15.3

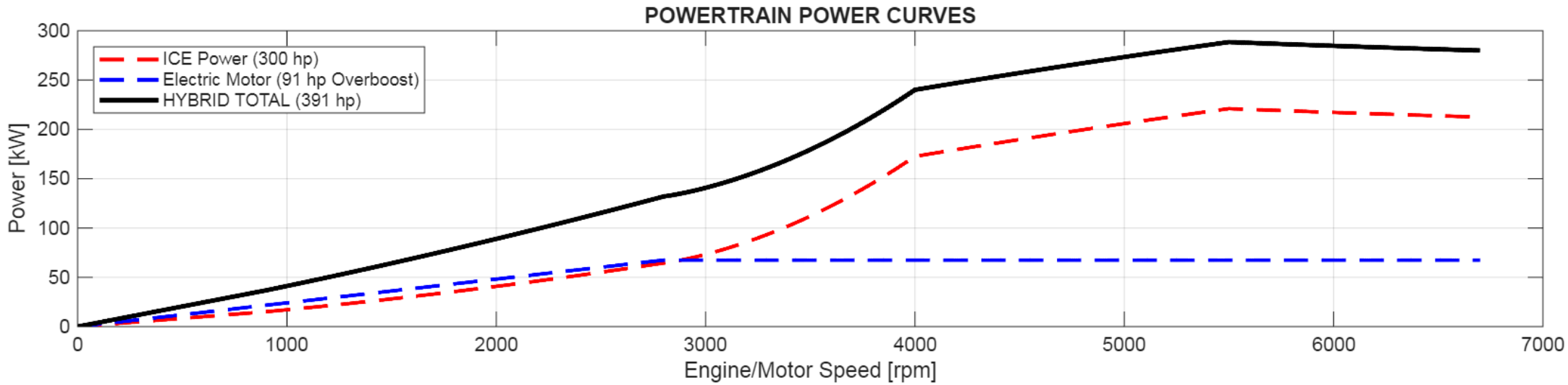


Hybrid mode	
Speed [Km/h]	Time [s]
100	7.7
200	18.1

Emode	
Speed [Km/h]	Time [s]
100	13.2
200	N/A

Overall Results – Turbo Lag Elimination





Overall Results – Final Recap

PERFORMANCE METRIC	VALUES
Electric driving mode	*~ 20 km in E-mode range*
Fuel consumption and emissions	*- 42%*
0-100 km/h	*5.9 s (- 2.3 s faster)*
0-200 km/h	*15.3 s (- 7.0 s faster)*
Imola racetrack lap time	*- 3.4 s (3% faster)*
Additional system weight	*+ 59 kg*
Additional conversion cost	+ 9% (€ 10340)

KEY ARCHITECTURAL HIGHLIGHTS:

- Mass Penalty Contained: Only +59kg core powertrain addition (turnkey prototype with K0 clutch, cabling and e-reverse deletion).
- Dynamics Rebalancing: Front battery installation shifts weight distribution from 39/61 to 41.5/58.5, mitigating historic snap-oversteer.
- Zero Aero / Packaging Compromise: Preserves original Whale Tail body, Fuchs wheels (205/55 R 16 & 245/45 R16), and lowers CoG to 0.445 m.

THANKS FOR YOUR ATTENTION



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